

SUDA-Muon: Decentralized Muon Optimization via a Unified Primal-Dual Framework

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Abstract. Fully decentralized Muon is difficult for a specific reason: the nonlinear matrix sign operator msgn does not commute with gossip averaging. As a result, extending Muon from centralized training to sparse peer-to-peer networks is not just a matter of adding communication; one must decide how polarization, tracking, and consensus correction should interact. We propose SUDA-Muon, which places Muon's matrix polarization directions inside the SUDA primal-dual backbone parameterized by (A, B, C) . The same template recovers ED/D², EXTRA, gradient tracking, and related decentralized schemes, so the communication mechanism can vary while the Muon geometry is kept fixed. We prove a topology-separated non-asymptotic rate in the nuclear-norm geometry: the leading term $\mathcal{O}((1 + \sigma/\sqrt{N})K^{-1/4})$ is independent of the graph, and topology appears only through faster-decaying corrections involving the SUDA contraction rate and the spectral gap. We also show by a smooth counterexample that tracking before polarization is not optional: removing it can trap the averaged iterate at a non-stationary point. We further give a linear-speedup counterexample that separates federated and fully decentralized Muon: server-side averaging before polarization enjoys the usual $\sigma/\sqrt{N}$ variance reduction, whereas local polarization before averaging does not, even under exact averaging on a complete graph. Experiments on CIFAR-100 and GPT-2 fine-tuning support the same qualitative picture. On short-horizon ring benchmarks, decentralized Muon variants are often close; in the 20-node, 100-epoch non-IID CIFAR-100 setting both SUDA-based variants outperform DEMUON, with SUDA-MUON-ED performing best, while under near-IID partitions the gap between different SUDA-Muon instances becomes much smaller and the preferred topology changes.

AMS subject classifications: 90C26, 90C25, 68W15

Key words: decentralized optimization, matrix-aware optimizer, Muon, topology separation, convergence analysis

1 Introduction

Fully decentralized training is attractive in settings where data and compute are naturally spread across agents and a central coordinator is either unavailable or undesirable [3, 4, 8]. In this paper, however, we are interested in a narrower question than decentralized

learning in general: how should one extend *Muon*, a matrix-structured optimizer, to a sparse peer-to-peer network? The main obstacle is that *Muon*'s matrix sign operator *msgn* is nonlinear, whereas decentralized communication is built from linear averaging. Once these two operations are combined, averaging and polarization need not agree, so the usual intuition from decentralized SGD no longer applies.

To tackle this problem, we consider the decentralized matrix optimization problem

$$\min_{X \in \mathbb{R}^{m \times n}} f(X) \triangleq \frac{1}{N} \sum_{i=1}^N f_i(X), \quad (1.1)$$

where agent $i \in [N] \triangleq \{1, \dots, N\}$ only has access to its local objective f_i and can query a stochastic first-order oracle. At iteration k , given a local iterate X_i^k , agent i samples ζ_i^k and obtains a stochastic gradient

$$G_i^k \triangleq \nabla F_i(X_i^k; \zeta_i^k) \in \mathbb{R}^{m \times n}, \quad (1.2)$$

which satisfies standard unbiasedness and bounded-variance conditions. The agents are connected by an undirected communication graph encoded by a symmetric, doubly stochastic mixing matrix $W \in \mathbb{R}^{N \times N}$, and they can exchange messages only with their neighbors according to W .

In this setting, decentralized optimization already offers several correction mechanisms for heterogeneity, including gradient tracking, EXTRA, and ED/D² [8, 13, 15]. SUDA is particularly appealing because it expresses many of these schemes inside one primal–dual recursion parameterized by three matrices (A, B, C) built from W [2]. On the optimizer side, *Muon* replaces a coordinatewise step by a matrix polarization direction and has shown strong empirical performance in centralized and federated settings [7, 9, 10, 14]. What is still missing is a framework that lets these two viewpoints meet cleanly: we would like to vary the communication backbone without redesigning the *Muon* update from scratch every time.

Current decentralized *Muon* results do not yet provide that picture. De*Muon* [6] combines *Muon* with gradient tracking and works over general graphs, but it is built around one specific communication template. This leaves four questions open. Can *Muon* be placed inside a unified SUDA-style backbone? Can one obtain a topology-separated rate in the nuclear-norm geometry induced by *Muon*, rather than an analysis in which network effects and optimizer geometry are entangled from the start? Is tracking before polarization genuinely necessary, or merely convenient? And, compared with server-based/federated *Muon*, can a fully decentralized *Muon* update still exploit the usual linear-speedup effect from averaging stochastic gradients?

We answer these questions with SUDA–*Muon*, which keeps *Muon*'s matrix geometry but places it inside the SUDA primal–dual recursion. The point is not only to obtain one more decentralized *Muon* method. It is to expose which parts of the behavior come from the communication mechanism and which parts come from the polarization step itself.

Our contributions are the following.

- We propose SUDA–Muon, a unified decentralized Muon framework in which the communication mechanism is encoded by the SUDA matrices (A, B, C) . By choosing these matrices appropriately, the same template recovers ED/D², EXTRA, and ATC-style gradient tracking while keeping the Muon direction fixed.
- We prove a topology-separated non-asymptotic convergence guarantee in the Muon-induced nuclear-norm geometry. The full five-term bound is stated in Corollary 4.1. In particular, with suitable stepsize and EMA schedules, its dominant term scales as $\mathcal{O}((1 + \sigma/\sqrt{N})K^{-1/4})$, while topology enters only through faster-decaying corrections involving the SUDA contraction rate γ and the mixing rate λ .
- We show that tracking before polarization is structurally necessary. Via a smooth matrix-logistic counterexample, we prove that if each node applies msgn directly to its local EMA, the averaged iterate can remain fixed at a non-stationary point.
- We give a second counterexample that isolates the difference between federated and fully decentralized Muon. In a simple noisy quadratic, server-side averaging before polarization yields the usual $\sigma/\sqrt{N}$ variance reduction, whereas local polarization before averaging produces an update whose contraction rate is independent of N , even under exact averaging on a complete graph.
- We evaluate the framework on CIFAR-100 classification and GPT-2 fine-tuning. The experiments support a more specific qualitative picture: short-horizon ring benchmarks do not strongly separate decentralized Muon variants, but in the 20-node, 100-epoch non-IID CIFAR-100 setting both SUDA-based variants outperform DEMUON, with SUDA–MUON-ED performing best; under near-IID partitions, the gap between different SUDA–Muon instances becomes much smaller and the preferred topology changes.

2 Related Work

2.1 Matrix-Aware Optimizers and the Muon Family in Centralized Training

Muon was introduced as an optimizer that acts directly on matrix-valued parameters through a polarization step instead of treating every layer as a flattened vector [7]. Subsequent empirical work reports strong performance on modern architectures, including large language models, and recent theory interprets Muon as steepest descent under a spectral-norm constraint [9, 12, 16]. For our purposes, the important point is not merely that Muon is matrix-aware, but that its update rule is nonlinear at the matrix level. That nonlinearity is exactly what complicates decentralization.

A broader literature studies structured optimizers such as Adafactor and Shampoo [5, 11]. These methods also exploit matrix or tensor structure, but they do not raise the same question considered here. Our focus is narrower: how to combine Muon’s polarization step with fully decentralized correction mechanisms over sparse graphs.

2.2 Decentralized Optimization and Heterogeneity Correction

Decentralized SGD is simple but biased under heterogeneous data. EXTRA, Exact Diffusion/ D^2 , and gradient tracking were developed to remove or reduce that bias through additional correction terms [13, 15, 17]. SUDA is especially relevant because it places several of these methods inside one primal–dual template parameterized by (A, B, C) , making it possible to compare communication mechanisms without redesigning the algorithm each time [1, 18].

Most of this literature, however, works in a Euclidean setting: gradients are communicated linearly, and the optimizer itself is vector-based or at most diagonally preconditioned. The resulting theory says a great deal about topology and bias correction, but it does not address what happens when the optimizer applies a nonlinear matrix transformation before the update.

2.3 Distributed Muon: Decentralized and Federated Extensions

Distributed extensions of Muon are already beginning to appear. Federated variants such as FedMuon study star-topology settings and show that naive local Muon combined with FedAvg can be unstable under heterogeneity, which motivates additional server-side correction [10, 14]. Those works are informative about heterogeneity, but they do not address fully decentralized communication.

DeMuon is, to our knowledge, the clearest fully decentralized Muon method on general graphs [6]. It demonstrates that combining Muon with gradient tracking is viable, but it is organized around one specific communication template. Our paper differs in emphasis: we ask for a unified SUDA-style backbone for Muon, a topology-separated rate within that backbone, a direct statement that tracking before polarization is not merely helpful but necessary in general, and an explicit counterexample separating the linear-speedup behavior of federated and fully decentralized Muon.

3 Problem Formulation and Algorithm

3.1 Problem Setup

We consider the decentralized matrix optimization problem

$$\min_{X \in \mathbb{R}^{m \times n}} f(X) \triangleq \frac{1}{N} \sum_{i=1}^N f_i(X),$$

where agent $i \in [N] \triangleq \{1, \dots, N\}$ can only access its local objective f_i and query a stochastic first-order oracle. At iteration k , given X_i^k , agent i samples ζ_i^k and obtains the stochastic gradient

$$G_i^k \triangleq \nabla F_i(X_i^k; \zeta_i^k) \in \mathbb{R}^{m \times n}.$$

Algorithm 1 SUDA–Muon (stacked form)

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1: Input: stepsize  $\alpha > 0$ , EMA parameter  $\beta \in [0, 1)$ , mixing / SUDA matrices  $(W, A, B, C)$ .
2: Initialize: choose  $\mathbf{X}^0 = \text{col}\{X_1^0, \dots, X_N^0\}$ , set  $\mathbf{Y}^0 = 0$ .
3: Query stochastic gradients  $\mathbf{G}^0 = \text{col}\{G_1^0, \dots, G_N^0\}$  at  $\mathbf{X}^0$ .
4: Set  $\mathbf{M}^0 := \mathbf{G}^0$  and  $\mathbf{H}^0 := \mathbf{M}^0$ .
5: for  $k = 0, 1, 2, \dots$  do
6:   Query stochastic gradients  $\mathbf{G}^k = \text{col}\{G_1^k, \dots, G_N^k\}$  at  $\mathbf{X}^k$ .
7:    $\mathbf{M}^{k+1} \leftarrow \beta \mathbf{M}^k + (1 - \beta) \mathbf{G}^k$  (EMA)
8:    $\mathbf{H}^{k+1} \leftarrow W(\mathbf{H}^k + \mathbf{M}^{k+1} - \mathbf{M}^k)$  (gradient tracking)
9:    $\mathbf{X}^{k+1} \leftarrow A(C\mathbf{X}^k - \alpha \text{msgn}(\mathbf{H}^{k+1})) - B\mathbf{Y}^k$  (SUDA backbone)
10:   $\mathbf{Y}^{k+1} \leftarrow \mathbf{Y}^k + B\mathbf{X}^{k+1}$ 
11: end for

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3.2 Algorithm Description

The framework is parameterized by the SUDA communication matrices (A, B, C) , which are low-degree polynomials of the mixing matrix W . Different choices recover well-known decentralized schemes as special cases:

- **ED/D²:** $A = C = W$ and $B^2 = I - W^2$.
- **EXTRA:** $A = C = \frac{I+W}{2}$ and $B^2 = \frac{I-W}{2}$.
- **Gradient Tracking (ATC-GT):** $A = C = W$ and $B = I - W$.

Remark 3.1. DeMuon [6] uses a closely related gradient-tracking mechanism combined with Muon directions, but applies mixing directly to the primal variables rather than through the SUDA primal–dual backbone. Specifically, DeMuon does not maintain the dual variable $\mathbf{Y}^k$ and instead performs a gossip-style averaging step on $\mathbf{X}^k$ after the Muon update. This subtle structural difference means DeMuon is not recovered as a special case of Algorithm 1, but its gradient-tracking component shares the same motivations analyzed in our convergence theory.

In all cases, the Muon polarization direction $\text{msgn}(\mathbf{H}^{k+1})$ is applied *after* gradient tracking, so that the msgn operator acts on a network-aggregated quantity rather than a purely local signal. The stacked update equations are:

$$\begin{aligned}
\mathbf{X}^{k+1} &= A(C\mathbf{X}^k - \alpha \text{msgn}(\mathbf{H}^{k+1})) - B\mathbf{Y}^k, \\
\mathbf{Y}^{k+1} &= \mathbf{Y}^k + B\mathbf{X}^{k+1}, \\
\mathbf{M}^{k+1} &= \beta \mathbf{M}^k + (1 - \beta) \mathbf{G}^k, \\
\mathbf{H}^{k+1} &= W(\mathbf{H}^k + \mathbf{M}^{k+1} - \mathbf{M}^k),
\end{aligned} \tag{3.1}$$

3.3 Assumptions

We adopt standard assumptions from refined SUDA analyses and recent Muon/DeMuon convergence analyses, adapted to matrix-valued parameters.

Assumption 1 (Network and SUDA communication structure). Let $W \in \mathbb{R}^{N \times N}$ be the (undirected) mixing matrix associated with the communication graph. We assume:

1. W is symmetric, doubly stochastic, and primitive. Denote the mixing rate

$$\lambda \triangleq \rho\left(W - \frac{1}{N}\mathbf{1}_N\mathbf{1}_N^\top\right) \in (0,1).$$

2. The SUDA matrices $A, B, C \in \mathbb{R}^{N \times N}$ satisfy: (i) A and C are doubly stochastic; (ii) $BZ = 0$ iff $Z_1 = \dots = Z_N$ for stacked variables; (iii) A , B^2 , and C are (low-degree) polynomial functions of W .

Assumption 2 (Matrix smoothness and lower boundedness). Each local objective $f_i : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ is L_* -smooth in the sense that for all $X, Y \in \mathbb{R}^{m \times n}$,

$$\|\nabla f_i(X) - \nabla f_i(Y)\|_* \leq L_* \|X - Y\|, \quad \forall i \in [N],$$

where $\|\cdot\|_*$ and $\|\cdot\|$ denote the nuclear norm and operator norm, respectively. Moreover, the global objective $f(X) \triangleq \frac{1}{N} \sum_{i=1}^N f_i(X)$ is bounded below: $f(X) \geq f_{\inf} > -\infty$.

Assumption 3 (Stochastic first-order oracle). At each iteration k and node i , the oracle returns $G_i^k = \nabla F_i(X_i^k; \zeta_i^k)$ satisfying

$$\mathbb{E}[G_i^k | \mathcal{F}^k] = \nabla f_i(X_i^k), \quad \mathbb{E}[\|G_i^k - \nabla f_i(X_i^k)\|_F^2 | \mathcal{F}^k] \leq \sigma^2,$$

where $\mathcal{F}^k$ is the natural filtration of the algorithm. Stochastic gradients are conditionally independent across agents and time given $\mathcal{F}^k$.

Assumption 4 (Muon polarization operator). For any $H \in \mathbb{R}^{m \times n}$ with a reduced SVD $H = U\Sigma V^\top$, define

$$\text{msgn}(H) \triangleq UV^\top, \quad \text{msgn}(0) \triangleq 0. \quad (3.2)$$

Each agent uses $S_i^{k+1} = \text{msgn}(H_i^{k+1})$ as the Muon direction, computed locally (e.g., via an exact SVD or approximate Newton–Schulz iterations).

4 Theoretical Analysis

We present the theoretical analysis of SUDA–Muon. For the reader’s convenience, we first state the main convergence result and its big- $\mathcal{O}$ simplification in Subsection 4.1, then establish the proof ingredients and final assembly in Subsection 4.3. Subsections 4.4 and 4.5 then draw two structural consequences of this theory: how the master bound specializes to concrete SUDA backbones, and why tracking must precede polarization.

4.1 Statement of Main Results

The following proposition provides the explicit convergence rate for SUDA–Muon.

Proposition 4.1 (Main convergence rate — informal statement). *Under Assumptions 1–4 and standard SUDA stability conditions, choose the stepsize $\alpha = \alpha_0 K^{-3/4}$ and EMA parameter $1 - \beta = b_0 K^{-1/2}$. Then the average nuclear-norm stationarity measure of SUDA–Muon satisfies*

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\bar{X}^k)\|_*] = \mathcal{O} \left(\underbrace{\left(1 + \frac{\sigma}{\sqrt{N}}\right) K^{-1/4}}_{\text{topology-free}} + \underbrace{\frac{K^{-3/4}}{(1-\gamma)(1-\lambda)}}_{\text{network-dependent}} + \text{lower-order terms} \right), \quad (4.1)$$

where $\gamma \in (0,1)$ is the SUDA contraction rate and $\lambda = \rho(W - J) \in (0,1)$ is the mixing rate.

The precise five-term bound, together with all constants, is given in Proposition 4.4 and its big- $\mathcal{O}$ simplification in Corollary 4.1. Three structural features of this result deserve emphasis:

1. **Topology separation.** The leading term $(1 + \sigma/\sqrt{N})K^{-1/4}$ is entirely free of any network quantity; the communication topology affects only the faster-decaying correction terms.
2. **Modular network dependence.** Network effects enter only through $(1-\gamma)$ and $(1-\lambda)$, and different choices of (A, B, C) alter only γ , so one can compare communication strategies by simply substituting the corresponding SUDA contraction rate (see Subsection 4.4).
3. **Muon-induced geometry.** The stationarity measure is the nuclear norm $\|\nabla f\|_*$, which is the natural dual norm for Muon’s operator-norm constraint set. This is fundamentally different from the Euclidean norm used in standard decentralized SGD analyses.

The proof proceeds in three stages: we first derive a one-step descent inequality in the nuclear-norm geometry, then establish the auxiliary bounds controlling EMA error, gradient tracking, SUDA disagreement, and forcing terms, and finally assemble these ingredients via telescoping and the chosen stepsize/EMA schedules.

4.2 Basic Properties of the Matrix Sign Operator

Lemma 4.1 (Basic properties of msgn and a robust inner-product lower bound). *Let $H \in \mathbb{R}^{m \times n}$ and define $\text{msgn}(H)$ as in (3.2): if $H \neq 0$ has reduced SVD $H = U\Sigma V^\top$ with $U \in \mathbb{R}^{m \times r}$, $V \in \mathbb{R}^{n \times r}$, $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_r)$ and $\sigma_i > 0$, then $\text{msgn}(H) := UV^\top$, and set $\text{msgn}(0) := 0$. Then the following hold.*

1. **Alignment with the nuclear norm.** One has

$$\langle H, \text{msgn}(H) \rangle = \|H\|_*. \quad (4.2)$$

2. **Operator- and Frobenius-norm bounds.** One has

$$\|\text{msgn}(H)\| \leq 1, \quad \|\text{msgn}(H)\|_F = \sqrt{\text{rank}(H)} \leq \sqrt{r_{\max}}, \quad r_{\max} := \min\{m, n\}. \quad (4.3)$$

Moreover, if $H \neq 0$ then $\|\text{msgn}(H)\| = 1$.

3. **A key inequality (no Lipschitzness required).** For any $G, H \in \mathbb{R}^{m \times n}$,

$$\langle G, \text{msgn}(H) \rangle \geq \|G\|_* - 2\|G - H\|_*. \quad (4.4)$$

The proof uses only the definition (3.2), basic linear algebra, and the standard nuclear/operator norm duality inequality

$$|\langle A, B \rangle| \leq \|A\|_* \|B\|, \quad \forall A, B \in \mathbb{R}^{m \times n}, \quad (4.5)$$

where $\langle A, B \rangle := \text{tr}(A^\top B)$.

Proof. If $H = 0$, then $\text{msgn}(H) = 0$ by definition and all claims are immediate. Hence assume $H \neq 0$ and write its reduced SVD as $H = U\Sigma V^\top$ with $U^\top U = I_r$ and $V^\top V = I_r$.

Part (1): $\langle H, \text{msgn}(H) \rangle = \|H\|_*$. Using cyclicity of the trace and $U^\top U = I_r$, $V^\top V = I_r$,

$$\langle H, \text{msgn}(H) \rangle = \text{tr}((U\Sigma V^\top)^\top (UV^\top)) = \text{tr}(V\Sigma U^\top UV^\top) = \text{tr}(V\Sigma V^\top) = \text{tr}(\Sigma) = \sum_{i=1}^r \sigma_i = \|H\|_*.$$

Part (2): operator- and Frobenius-norm bounds. Set $M := \text{msgn}(H) = UV^\top$. Then

$$M^\top M = VU^\top UV^\top = VV^\top,$$

which is the orthogonal projector onto $\text{col}(V)$ and hence has eigenvalues in $\{0, 1\}$. Therefore

$$\|M\|^2 = \lambda_{\max}(M^\top M) = \lambda_{\max}(VV^\top) = 1,$$

so $\|\text{msgn}(H)\| = 1$ for $H \neq 0$, and the bound $\|\text{msgn}(H)\| \leq 1$ holds in general (including $H = 0$). Moreover,

$$\|M\|_F^2 = \text{tr}(M^\top M) = \text{tr}(VV^\top) = \text{rank}(VV^\top) = r = \text{rank}(H),$$

so $\|\text{msgn}(H)\|_F = \sqrt{\text{rank}(H)} \leq \sqrt{\min\{m, n\}} =: \sqrt{r_{\max}}$.

Part (3): robust inner-product lower bound. Write

$$\langle G, \text{msgn}(H) \rangle = \langle H, \text{msgn}(H) \rangle + \langle G - H, \text{msgn}(H) \rangle.$$

By Part (1), $\langle H, \text{msgn}(H) \rangle = \|H\|_*$. For the second term, apply the duality inequality (4.5) with $A = G - H$ and $B = \text{msgn}(H)$, together with Part (2) ($\|\text{msgn}(H)\| \leq 1$), to obtain

$$\langle G - H, \text{msgn}(H) \rangle \geq -|\langle G - H, \text{msgn}(H) \rangle| \geq -\|G - H\|_* \|\text{msgn}(H)\| \geq -\|G - H\|_*.$$

Consequently,

$$\langle G, \text{msgn}(H) \rangle \geq \|H\|_* - \|G - H\|_*. \quad (4.6)$$

Finally, by the triangle inequality for the nuclear norm,

$$\|G\|_* = \|H + (G - H)\|_* \leq \|H\|_* + \|G - H\|_*,$$

which rearranges to $\|H\|_* \geq \|G\|_* - \|G - H\|_*$. Plugging this into (4.6) yields

$$\langle G, \text{msgn}(H) \rangle \geq (\|G\|_* - \|G - H\|_*) - \|G - H\|_* = \|G\|_* - 2\|G - H\|_*,$$

which is (4.4). $\square$

4.3 Proof Ingredients and Main Proofs

This subsection develops the technical ingredients behind the master rate and then assembles them into the final convergence bound.

One-Step Descent Analysis

Lemma 4.2 (Averaged SUDA–Muon dynamics). *Suppose Assumption 1 holds and the iterates are generated by (3.1) with initialization $Y_i^0 = 0$ for all $i \in [N]$. Define the averages*

$$\bar{X}^k \triangleq \frac{1}{N} \sum_{i=1}^N X_i^k, \quad \bar{S}^{k+1} \triangleq \frac{1}{N} \sum_{i=1}^N S_i^{k+1}, \quad S_i^{k+1} = \text{msgn}(H_i^{k+1}).$$

Then, for all $k \geq 0$,

$$\bar{X}^{k+1} = \bar{X}^k - \alpha \bar{S}^{k+1}. \quad (4.7)$$

Proof. Let $\mathbf{1} \in \mathbb{R}^N$ denote the all-ones vector and set

$$J \triangleq \frac{1}{N} \mathbf{1} \mathbf{1}^\top \in \mathbb{R}^{N \times N}.$$

For a stacked variable $\mathbf{Z} = \text{col}\{Z_1, \dots, Z_N\}$ (with matrix blocks $Z_i \in \mathbb{R}^{m \times n}$), $J\mathbf{Z}$ has all blocks equal to the average $\bar{Z} \triangleq \frac{1}{N} \sum_i Z_i$.

From (3.1),

$$\mathbf{X}^{k+1} = A \left(C\mathbf{X}^k - \alpha \text{msgn}(\mathbf{H}^{k+1}) \right) - B\mathbf{Y}^k.$$

Left-multiplying by J and using that A and C are doubly stochastic (Assumption 1(2)(i)), hence $JA = J$ and $JC = J$, yields

$$J\mathbf{X}^{k+1} = J\mathbf{X}^k - \alpha J \text{msgn}(\mathbf{H}^{k+1}) - JB\mathbf{Y}^k. \quad (4.8)$$

We claim that $JBY^k=0$ for all $k \geq 0$ under $\mathbf{Y}^0=0$. Indeed, define the row vector $b^\top \triangleq \frac{1}{N} \mathbf{1}^\top B$ and the stacked linear functional $u^k \triangleq b^\top \mathbf{Y}^k \in \mathbb{R}^{m \times n}$. From the dual update $\mathbf{Y}^{k+1} = \mathbf{Y}^k + B\mathbf{X}^{k+1}$,

$$u^{k+1} = u^k + b^\top B\mathbf{X}^{k+1} = u^k + \frac{1}{N} \mathbf{1}^\top B^2 \mathbf{X}^{k+1}.$$

By Assumption 1(2)(ii), for every *consensus* stacked variable $\mathbf{Z} = \text{col}\{Z, \dots, Z\}$ (with an arbitrary block $Z \in \mathbb{R}^{m \times n}$), we have $B\mathbf{Z}=0$. Writing the i -th block of $B\mathbf{Z}$ as $(B\mathbf{Z})_i = \sum_{j=1}^N b_{ij}Z$, we obtain $(\sum_{j=1}^N b_{ij})Z = 0$ for every Z . Choosing any nonzero Z (e.g., a basis matrix) implies $\sum_{j=1}^N b_{ij} = 0$ for all i , i.e.,

$$B\mathbf{1} = 0. \quad (4.9)$$

Consequently, $B^2\mathbf{1} = 0$.

Next, by Assumption 1(2)(iii), there exists a polynomial p such that $B^2 = p(W)$. Since W is doubly stochastic, $W\mathbf{1} = \mathbf{1}$ and $\mathbf{1}^\top W = \mathbf{1}^\top$, hence $p(W)\mathbf{1} = p(1)\mathbf{1}$ and $\mathbf{1}^\top p(W) = p(1)\mathbf{1}^\top$. From $B^2\mathbf{1} = p(W)\mathbf{1} = 0$ we conclude $p(1) = 0$, and therefore

$$\mathbf{1}^\top B^2 = \mathbf{1}^\top p(W) = p(1)\mathbf{1}^\top = 0. \quad (4.10)$$

Therefore $b^\top B = \frac{1}{N} \mathbf{1}^\top B^2 = 0$, and thus $u^{k+1} = u^k$. Since $\mathbf{Y}^0 = 0$, we get $u^0 = 0$ and hence $u^k \equiv 0$ for all k . Finally,

$$JBY^k = \frac{1}{N} \mathbf{1} (\mathbf{1}^\top B\mathbf{Y}^k) = \mathbf{1} u^k = 0.$$

Substituting $JBY^k = 0$ into (4.8) and taking the common block yields $\bar{X}^{k+1} = \bar{X}^k - \alpha \bar{S}^{k+1}$, which is (4.7). $\square$

Lemma 4.3 (One-step descent of $f(\bar{X}^k)$ with Muon direction and tracking error). *Suppose Assumptions 1 and 2 hold, the iterates are generated by (3.1), and $Y_i^0 = 0$ for all $i \in [N]$ (as in Algorithm 1). Let $(\bar{X}^k)_{k \geq 0}$ be the averaged iterates defined in Lemma 4.2. Then, for all $k \geq 0$,*

$$f(\bar{X}^{k+1}) \leq f(\bar{X}^k) - \alpha \|\nabla f(\bar{X}^k)\|_* + \frac{L_* \alpha^2}{2} + \frac{2\alpha}{N} \sum_{i=1}^N \|\nabla f(\bar{X}^k) - H_i^{k+1}\|_*. \quad (4.11)$$

Here and throughout, $\langle A, B \rangle := \text{tr}(A^\top B)$ denotes the matrix inner product.

Proof. Step 1: a descent lemma under Assumption 2. Since $f(X) = \frac{1}{N} \sum_{i=1}^N f_i(X)$, Assumption 2 implies that for all X, Y ,

$$\|\nabla f(X) - \nabla f(Y)\|_* = \left\| \frac{1}{N} \sum_{i=1}^N (\nabla f_i(X) - \nabla f_i(Y)) \right\|_* \leq \frac{1}{N} \sum_{i=1}^N \|\nabla f_i(X) - \nabla f_i(Y)\|_* \leq L_* \|X - Y\|. \quad (4.12)$$

Fix $X, Y \in \mathbb{R}^{m \times n}$ and let $D := Y - X$. Define $g(t) := f(X + tD)$ for $t \in [0, 1]$. Then g is differentiable and

$$g'(t) = \langle \nabla f(X + tD), D \rangle.$$

Using the fundamental theorem of calculus,

$$f(Y) - f(X) = g(1) - g(0) = \int_0^1 \langle \nabla f(X + tD), D \rangle dt. \quad (4.13)$$

Add and subtract $\nabla f(X)$ inside the integrand to obtain

$$f(Y) - f(X) = \langle \nabla f(X), D \rangle + \int_0^1 \langle \nabla f(X + tD) - \nabla f(X), D \rangle dt.$$

By the nuclear/operator duality inequality $|\langle A, B \rangle| \leq \|A\|_* \|B\|$ and (4.12), for each $t \in [0, 1]$,

$$\begin{aligned} \langle \nabla f(X + tD) - \nabla f(X), D \rangle &\leq |\langle \nabla f(X + tD) - \nabla f(X), D \rangle| \\ &\leq \|\nabla f(X + tD) - \nabla f(X)\|_* \|D\| \\ &\leq L_* \|tD\| \|D\| = L_* t \|D\|^2. \end{aligned}$$

Plugging this into (4.13) gives the descent inequality

$$f(Y) \leq f(X) + \langle \nabla f(X), Y - X \rangle + \frac{L_*}{2} \|Y - X\|^2. \quad (4.14)$$

Step 2: apply (4.14) to the averaged recursion. By Lemma 4.2, $\bar{X}^{k+1} = \bar{X}^k - \alpha \bar{S}^{k+1}$ with $\bar{S}^{k+1} = \frac{1}{N} \sum_{i=1}^N S_i^{k+1}$ and $S_i^{k+1} = \text{msgn}(H_i^{k+1})$. Applying (4.14) with $X = \bar{X}^k$ and $Y = \bar{X}^{k+1}$ yields

$$f(\bar{X}^{k+1}) \leq f(\bar{X}^k) - \alpha \langle \nabla f(\bar{X}^k), \bar{S}^{k+1} \rangle + \frac{L_* \alpha^2}{2} \|\bar{S}^{k+1}\|^2. \quad (4.15)$$

Moreover, by Lemma 4.1(2), $\|S_i^{k+1}\| \leq 1$ for all i , and by convexity of the operator norm,

$$\|\bar{S}^{k+1}\| = \left\| \frac{1}{N} \sum_{i=1}^N S_i^{k+1} \right\| \leq \frac{1}{N} \sum_{i=1}^N \|S_i^{k+1}\| \leq 1. \quad (4.16)$$

Combining (4.15) and (4.16) gives

$$f(\bar{X}^{k+1}) \leq f(\bar{X}^k) - \alpha \langle \nabla f(\bar{X}^k), \bar{S}^{k+1} \rangle + \frac{L_* \alpha^2}{2}. \quad (4.17)$$

Step 3: lower bound $\langle \nabla f(\bar{X}^k), \bar{S}^{k+1} \rangle$ via Lemma 4.1. Let $G := \nabla f(\bar{X}^k)$. For each agent i , Lemma 4.1(3) with $H = H_i^{k+1}$ gives

$$\langle G, S_i^{k+1} \rangle = \langle G, \text{msgn}(H_i^{k+1}) \rangle \geq \|G\|_* - 2\|G - H_i^{k+1}\|_*.$$

Averaging over i and using $\langle G, \bar{S}^{k+1} \rangle = \frac{1}{N} \sum_{i=1}^N \langle G, S_i^{k+1} \rangle$ yields

$$\langle \nabla f(\bar{X}^k), \bar{S}^{k+1} \rangle \geq \|\nabla f(\bar{X}^k)\|_* - \frac{2}{N} \sum_{i=1}^N \|\nabla f(\bar{X}^k) - H_i^{k+1}\|_*. \quad (4.18)$$

Substituting (4.18) into (4.17) gives (4.11). $\square$

Proposition 4.2 (Descent inequality with an explicit “tracking-error” remainder). *Suppose the assumptions of Lemma 4.3 hold. Then for any integer $K \geq 1$,*

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\bar{X}^k)\|_*] \leq \frac{\Delta_0}{\alpha K} + \frac{L_* \alpha}{2} + \frac{2}{NK} \sum_{k=0}^{K-1} \sum_{i=1}^N \mathbb{E} [\|\nabla f(\bar{X}^k) - H_i^{k+1}\|_*], \quad (4.19)$$

where $\Delta_0 := \mathbb{E}[f(\bar{X}^0)] - f_{\inf}$.

Proof. Sum the one-step inequality (4.11) from Lemma 4.3 over $k=0, \dots, K-1$, take expectations, and use $\mathbb{E}[f(\bar{X}^K)] \geq f_{\inf}$. Divide by αK . $\square$

EMA and Gradient Tracking Foundations

Lemma 4.4 (EMA stochastic error (per-agent and averaged)). *Suppose Assumption 3 holds and the iterates are generated by (3.1) (equivalently, Algorithm 1). Fix any agent $i \in [N]$. Define $C_i^0 := \nabla f_i(X_i^0)$ and for $k \geq 0$,*

$$C_i^{k+1} := \beta C_i^k + (1-\beta) \nabla f_i(X_i^k),$$

and recall that $M_i^0 = G_i^0$ and $M_i^{k+1} = \beta M_i^k + (1-\beta) G_i^k$. Then, for all $k \geq 0$,

$$\mathbb{E} \|C_i^k - M_i^k\|_F \leq \sqrt{\frac{1-\beta}{1+\beta}} \sigma + \beta^k \sigma. \quad (4.20)$$

Moreover, defining the averages $\bar{M}^k := \frac{1}{N} \sum_{i=1}^N M_i^k$ and $\bar{C}^k := \frac{1}{N} \sum_{i=1}^N C_i^k$, we have for all $k \geq 0$,

$$\mathbb{E} \|\bar{C}^k - \bar{M}^k\|_F \leq \sqrt{\frac{1-\beta}{1+\beta}} \frac{\sigma}{\sqrt{N}} + \beta^k \frac{\sigma}{\sqrt{N}}. \quad (4.21)$$

Finally, letting $r_{\max} := \min\{m, n\}$, for all $k \geq 0$,

$$\mathbb{E} \|C_i^k - M_i^k\|_* \leq \sqrt{r_{\max}} \mathbb{E} \|C_i^k - M_i^k\|_F. \quad (4.22)$$

Proof. Throughout, define the stochastic gradient noise

$$\Delta_i^k := G_i^k - \nabla f_i(X_i^k).$$

Then Assumption 3 implies

$$\mathbb{E} [\Delta_i^k | \mathcal{F}^k] = 0, \quad \mathbb{E} [\|\Delta_i^k\|_F^2 | \mathcal{F}^k] \leq \sigma^2. \quad (4.23)$$

Step 1: per-agent EMA error. Let $E_i^k := C_i^k - M_i^k$. By the definitions of C_i^{k+1} and M_i^{k+1} ,

$$E_i^{k+1} = \beta E_i^k + (1-\beta) (\nabla f_i(X_i^k) - G_i^k) = \beta E_i^k - (1-\beta) \Delta_i^k. \quad (4.24)$$

Unrolling (4.24) gives, for any $k \geq 0$,

$$E_i^k = (1-\beta) \sum_{t=0}^{k-1} \beta^{k-1-t} (\nabla f_i(X_i^t) - G_i^t) + \beta^k (\nabla f_i(X_i^0) - G_i^0). \quad (4.25)$$

Taking Frobenius norms, using the triangle inequality and Cauchy–Schwarz,

$$\begin{aligned} \mathbb{E} \|E_i^k\|_F &\leq (1-\beta) \mathbb{E} \left\| \sum_{t=0}^{k-1} \beta^{k-1-t} (\nabla f_i(X_i^t) - G_i^t) \right\|_F + \beta^k \mathbb{E} \|\nabla f_i(X_i^0) - G_i^0\|_F \\ &\leq (1-\beta) \sqrt{\mathbb{E} \left\| \sum_{t=0}^{k-1} \beta^{k-1-t} (\nabla f_i(X_i^t) - G_i^t) \right\|_F^2} + \beta^k \sqrt{\mathbb{E} \|\nabla f_i(X_i^0) - G_i^0\|_F^2}. \end{aligned} \quad (4.26)$$

The second term is bounded by $\beta^k \sigma$ using (4.23). For the first term, note that for $s < t$, by the tower property and (4.23),

$$\begin{aligned} &\mathbb{E} [\langle \nabla f_i(X_i^s) - G_i^s, \nabla f_i(X_i^t) - G_i^t \rangle] \\ &= \mathbb{E} \left[\mathbb{E} [\langle \nabla f_i(X_i^s) - G_i^s, \nabla f_i(X_i^t) - G_i^t \rangle | \mathcal{F}^t] \right] \\ &= \mathbb{E} \left[\langle \nabla f_i(X_i^s) - G_i^s, \underbrace{\mathbb{E} [\nabla f_i(X_i^t) - G_i^t | \mathcal{F}^t]}_{=0} \rangle \right] = 0. \end{aligned}$$

Hence all cross terms vanish in the expansion of the squared norm, and therefore

$$\begin{aligned} \mathbb{E} \left\| \sum_{t=0}^{k-1} \beta^{k-1-t} (\nabla f_i(X_i^t) - G_i^t) \right\|_F^2 &= \sum_{t=0}^{k-1} \beta^{2(k-1-t)} \mathbb{E} \|\nabla f_i(X_i^t) - G_i^t\|_F^2 \\ &\leq \sigma^2 \sum_{t=0}^{k-1} \beta^{2(k-1-t)} = \sigma^2 \frac{1-\beta^{2k}}{1-\beta^2} \leq \frac{\sigma^2}{1-\beta^2}. \end{aligned}$$

Plugging this into (4.26) yields

$$\mathbb{E} \|E_i^k\|_F \leq (1-\beta) \frac{\sigma}{\sqrt{1-\beta^2}} + \beta^k \sigma = \sqrt{\frac{1-\beta}{1+\beta}} \sigma + \beta^k \sigma,$$

which proves (4.20).

Step 2: averaged EMA error and the $1/\sqrt{N}$ factor. Let $\bar{E}^k := \bar{C}^k - \bar{M}^k = \frac{1}{N} \sum_{i=1}^N E_i^k$ and $\bar{\Delta}^k := \frac{1}{N} \sum_{i=1}^N \Delta_i^k$. Averaging (4.24) over i gives

$$\bar{E}^{k+1} = \beta \bar{E}^k - (1-\beta) \bar{\Delta}^k. \quad (4.27)$$

We now bound the second moment of $\bar{\Delta}^k$. By Assumption 3, conditional on $\mathcal{F}^k$ the random matrices $\{\Delta_i^k\}_{i=1}^N$ are independent across agents and satisfy (4.23). Therefore, conditional on $\mathcal{F}^k$,

$$\begin{aligned} \mathbb{E} [\|\bar{\Delta}^k\|_F^2 | \mathcal{F}^k] &= \mathbb{E} \left[\left\| \frac{1}{N} \sum_{i=1}^N \Delta_i^k \right\|_F^2 \middle| \mathcal{F}^k \right] = \frac{1}{N^2} \sum_{i=1}^N \mathbb{E} [\|\Delta_i^k\|_F^2 | \mathcal{F}^k] \\ &\leq \frac{\sigma^2}{N}. \end{aligned} \quad (4.28)$$

Equality (4.28) is exactly where the conditional independence across agents is used: it removes the cross terms $\langle \Delta_i^k, \Delta_j^k \rangle$ for $i \neq j$. Consequently, by Jensen,

$$\mathbb{E} \|\bar{\Delta}^k\|_F \leq \sqrt{\mathbb{E} \|\bar{\Delta}^k\|_F^2} \leq \frac{\sigma}{\sqrt{N}}. \quad (4.29)$$

Now apply the same argument as in Step 1 to the recursion (4.27), using (4.28) (in place of (4.23) with σ^2), to obtain

$$\mathbb{E} \|\bar{E}^k\|_F \leq \sqrt{\frac{1-\beta}{1+\beta}} \frac{\sigma}{\sqrt{N}} + \beta^k \frac{\sigma}{\sqrt{N}},$$

which is (4.21).

Step 3: nuclear norm conversion. For any matrix $A \in \mathbb{R}^{m \times n}$ with singular values $(s_j)_{j=1}^{r_{\max}}$,

$$\|A\|_* = \sum_{j=1}^{r_{\max}} s_j \leq \sqrt{r_{\max}} \left(\sum_{j=1}^{r_{\max}} s_j^2 \right)^{1/2} = \sqrt{r_{\max}} \|A\|_F,$$

where the inequality is Cauchy–Schwarz. Applying this pointwise with $A = E_i^k$ and taking expectations gives (4.22). $\square$

Lemma 4.5 (Gradient tracking preserves network averages). *Suppose Assumption 1(1) holds (in particular, W is doubly stochastic) and the gradient-tracking variable is updated as in (3.1):*

$$\mathbf{H}^{k+1} = W(\mathbf{H}^k + \mathbf{M}^{k+1} - \mathbf{M}^k), \quad k \geq 0.$$

Assume the initialization satisfies $H_i^0 = M_i^0$ for all $i \in [N]$ (equivalently, $\mathbf{H}^0 = \mathbf{M}^0$). Define the network averages

$$\bar{H}^k \triangleq \frac{1}{N} \sum_{i=1}^N H_i^k, \quad \bar{M}^k \triangleq \frac{1}{N} \sum_{i=1}^N M_i^k.$$

Then, for all $k \geq 0$,

$$\bar{H}^k = \bar{M}^k.$$

In particular, for all $k \geq 0$,

$$\bar{H}^{k+1} = \bar{H}^k + (\bar{M}^{k+1} - \bar{M}^k).$$

Proof. Let $\mathbf{1} \in \mathbb{R}^N$ denote the all-ones vector and set

$$J \triangleq \frac{1}{N} \mathbf{1} \mathbf{1}^\top \in \mathbb{R}^{N \times N}.$$

For any stacked variable $\mathbf{Z} = \text{col}\{Z_1, \dots, Z_N\}$ (with matrix blocks $Z_i \in \mathbb{R}^{m \times n}$), the stacked vector $J\mathbf{Z}$ has all blocks equal to the average $\bar{Z} \triangleq \frac{1}{N} \sum_i Z_i$.

Since W is doubly stochastic, we have $\mathbf{1}^\top W = \mathbf{1}^\top$, and thus

$$JW = \frac{1}{N} \mathbf{1}(\mathbf{1}^\top W) = \frac{1}{N} \mathbf{1}\mathbf{1}^\top = J.$$

Left-multiplying the gradient-tracking recursion by J and using $JW = J$ yields

$$\begin{aligned} J\mathbf{H}^{k+1} &= JW(\mathbf{H}^k + \mathbf{M}^{k+1} - \mathbf{M}^k) = J(\mathbf{H}^k + \mathbf{M}^{k+1} - \mathbf{M}^k) \\ &= J\mathbf{H}^k + (J\mathbf{M}^{k+1} - J\mathbf{M}^k), \quad \forall k \geq 0. \end{aligned}$$

Taking the common block of both sides (equivalently, premultiplying by $\frac{1}{N}\mathbf{1}^\top$) gives

$$\bar{H}^{k+1} = \bar{H}^k + (\bar{M}^{k+1} - \bar{M}^k), \quad \forall k \geq 0.$$

By the initialization $\bar{H}^0 = \bar{M}^0$, the above identity telescopes to

$$\bar{H}^k - \bar{M}^k = (\bar{H}^0 - \bar{M}^0) = 0,$$

for every $k \geq 0$. Hence $\bar{H}^k = \bar{M}^k$ for all $k \geq 0$. □

Lemma 4.6 (Gradient-tracking disagreement recursion). *Suppose Assumption 1(1) holds and the gradient-tracking variable obeys*

$$\mathbf{H}^{k+1} = W(\mathbf{H}^k + \mathbf{M}^{k+1} - \mathbf{M}^k), \quad k \geq 0.$$

Let $\mathbf{1}_N \in \mathbb{R}^N$ be the all-ones vector and define

$$J \triangleq \frac{1}{N} \mathbf{1}_N \mathbf{1}_N^\top \in \mathbb{R}^{N \times N}, \quad \tilde{\mathbf{Z}}^k \triangleq (I - J)\mathbf{Z}^k = \mathbf{Z}^k - J\mathbf{Z}^k.$$

Then, for all $k \geq 0$,

$$\|\tilde{\mathbf{H}}^{k+1}\|_F \leq \lambda \|\tilde{\mathbf{H}}^k + (\tilde{\mathbf{M}}^{k+1} - \tilde{\mathbf{M}}^k)\|_F,$$

where $\lambda \triangleq \rho(W - J) \in (0, 1)$ is the mixing rate from Assumption 1(1). Consequently, for any $\eta > 0$,

$$\|\tilde{\mathbf{H}}^{k+1}\|_F^2 \leq (1 + \eta)\lambda^2 \|\tilde{\mathbf{H}}^k\|_F^2 + (1 + 1/\eta)\lambda^2 \|\tilde{\mathbf{M}}^{k+1} - \tilde{\mathbf{M}}^k\|_F^2.$$

Proof. Since W is doubly stochastic, we have $W\mathbf{1}_N = \mathbf{1}_N$ and $\mathbf{1}_N^\top W = \mathbf{1}_N^\top$, which imply

$$WJ = \frac{1}{N} W\mathbf{1}_N \mathbf{1}_N^\top = \frac{1}{N} \mathbf{1}_N \mathbf{1}_N^\top = J, \quad JW = \frac{1}{N} \mathbf{1}_N (\mathbf{1}_N^\top W) = \frac{1}{N} \mathbf{1}_N \mathbf{1}_N^\top = J.$$

Therefore,

$$(I - J)W = W - JW = W - J.$$

Left-multiplying the gradient-tracking recursion by $(I - J)$ gives

$$\tilde{\mathbf{H}}^{k+1} = (I - J)\mathbf{H}^{k+1} = (I - J)W(\mathbf{H}^k + \mathbf{M}^{k+1} - \mathbf{M}^k) = (W - J)(\mathbf{H}^k + \mathbf{M}^{k+1} - \mathbf{M}^k).$$

Moreover, for any stacked variable $\mathbf{Z}$ we can decompose $\mathbf{Z} = J\mathbf{Z} + (I - J)\mathbf{Z}$, and since $(W - J)J = WJ - J^2 = J - J = 0$, we have

$$(W - J)\mathbf{Z} = (W - J)(I - J)\mathbf{Z} = (W - J)\tilde{\mathbf{Z}}.$$

Applying this identity to $\mathbf{H}^k$ and $\mathbf{M}^{k+1} - \mathbf{M}^k$ yields

$$\tilde{\mathbf{H}}^{k+1} = (W - J)(\tilde{\mathbf{H}}^k + (\tilde{\mathbf{M}}^{k+1} - \tilde{\mathbf{M}}^k)).$$

Next, view a stacked variable $\mathbf{Z} = \text{col}\{Z_1, \dots, Z_N\}$ (with matrix blocks $Z_i \in \mathbb{R}^{m \times n}$) as a vector by vectorization. Then the action of $W - J$ on the agent index corresponds to the linear map $((W - J) \otimes I_{mn})$ on $\text{vec}(\mathbf{Z})$, and

$$\|(W - J)\mathbf{Z}\|_F = \|((W - J) \otimes I_{mn})\text{vec}(\mathbf{Z})\|_2 \leq \|(W - J) \otimes I_{mn}\|_2 \|\text{vec}(\mathbf{Z})\|_2 = \|W - J\|_2 \|\mathbf{Z}\|_F.$$

Because W is symmetric, $W - J$ is also symmetric. Hence its spectral norm equals its spectral radius:

$$\|W - J\|_2 = \max_i |\lambda_i(W - J)| = \rho(W - J) = \lambda.$$

Combining the last two displays gives

$$\|\tilde{\mathbf{H}}^{k+1}\|_F \leq \lambda \|\tilde{\mathbf{H}}^k + (\tilde{\mathbf{M}}^{k+1} - \tilde{\mathbf{M}}^k)\|_F.$$

Finally, squaring and using the standard inequality $\|U + V\|_F^2 \leq (1 + \eta)\|U\|_F^2 + (1 + 1/\eta)\|V\|_F^2$ (valid for any $\eta > 0$) with $U = \tilde{\mathbf{H}}^k$ and $V = \tilde{\mathbf{M}}^{k+1} - \tilde{\mathbf{M}}^k$ yields

$$\begin{aligned} \|\tilde{\mathbf{H}}^{k+1}\|_F^2 &\leq \lambda^2 \|\tilde{\mathbf{H}}^k + (\tilde{\mathbf{M}}^{k+1} - \tilde{\mathbf{M}}^k)\|_F^2 \\ &\leq (1 + \eta)\lambda^2 \|\tilde{\mathbf{H}}^k\|_F^2 + (1 + 1/\eta)\lambda^2 \|\tilde{\mathbf{M}}^{k+1} - \tilde{\mathbf{M}}^k\|_F^2. \end{aligned}$$

This proves the claim. $\square$

SUDA Backbone Analysis

Lemma 4.7 (SUDA backbone disagreement bound with bounded input directions). *Suppose Assumption 1 holds. Consider the primal–dual recursion (stacked form)*

$$\begin{aligned} \mathbf{X}^{k+1} &= A(\mathbf{C}\mathbf{X}^k - \alpha \mathbf{U}^{k+1}) - B\mathbf{Y}^k, \\ \mathbf{Y}^{k+1} &= \mathbf{Y}^k + B\mathbf{X}^{k+1}, \end{aligned} \tag{4.30}$$

where $\alpha > 0$ and $\mathbf{U}^{k+1} = \text{col}\{U_1^{k+1}, \dots, U_N^{k+1}\}$ is an arbitrary (possibly time-varying) stacked input. Let $J := \frac{1}{N}\mathbf{1}\mathbf{1}^\top$ and $\tilde{\mathbf{Z}} := (I - J)\mathbf{Z}$. Assume $\mathbf{Y}^0 = 0$.

Assume additionally that the corresponding SUDA deviation matrix (defined in the proof) is stable, i.e., all of its eigenvalues are strictly inside the unit circle. Then there exist network-dependent constants

$$\gamma \in (0, 1), \quad v_1 \geq 1, \quad v_2 \geq 1, \quad \lambda_a \geq 0,$$

(depending only on A, B, C and the network) such that for all $k \geq 0$,

$$\begin{aligned} \|\tilde{\mathbf{X}}^k\|_F &\leq v_1 v_2 \gamma^k \left(\|\tilde{\mathbf{X}}^0\|_F + \|B^2\| \|\tilde{\mathbf{X}}^0\|_F \right) \\ &\quad + \frac{\alpha v_1 v_2 \lambda_a}{1-\gamma} \sup_{0 \leq t \leq k} \|\tilde{\mathbf{U}}^{t+1}\|_F. \end{aligned} \quad (4.31)$$

If, moreover, $\mathbf{U}^{k+1} = \text{msgn}(\mathbf{H}^{k+1})$ is the blockwise matrix-sign (Muon) direction from (3.1), then Lemma 4.1 implies

$$\|\tilde{\mathbf{U}}^{k+1}\|_F \leq 2\sqrt{Nr_{\max}}, \quad r_{\max} := \min\{m, n\}, \quad (4.32)$$

and thus

$$\|\tilde{\mathbf{X}}^k\|_F \leq v_1 v_2 \gamma^k (1 + \|B^2\|) \|\tilde{\mathbf{X}}^0\|_F + \frac{2\alpha v_1 v_2 \lambda_a}{1-\gamma} \sqrt{Nr_{\max}}.$$

Proof. Step 1: a non-incremental form involving B^2 only. Define

$$\mathbf{z}^k \triangleq \mathbf{Y}^k - B\mathbf{X}^k, \quad \mathbf{s}^k \triangleq B\mathbf{z}^k = B\mathbf{Y}^k - B^2\mathbf{X}^k.$$

Substituting $\mathbf{Y}^k = \mathbf{z}^k + B\mathbf{X}^k$ into (4.30) yields

$$\mathbf{X}^{k+1} = (AC - B^2)\mathbf{X}^k - \alpha A\mathbf{U}^{k+1} - B\mathbf{z}^k, \quad (4.33a)$$

$$\mathbf{z}^{k+1} = \mathbf{z}^k + B\mathbf{X}^k. \quad (4.33b)$$

Multiplying (4.33b) by B and using the definition of $\mathbf{s}^k$ gives

$$\mathbf{X}^{k+1} = (AC - B^2)\mathbf{X}^k - \alpha A\mathbf{U}^{k+1} - \mathbf{s}^k, \quad (4.34a)$$

$$\mathbf{s}^{k+1} = \mathbf{s}^k + B^2\mathbf{X}^k. \quad (4.34b)$$

Step 2: restriction to the disagreement subspace. Let $\tilde{\mathbf{Z}} := (I - J)\mathbf{Z}$. Since A, C, B^2 are polynomials in W (Assumption 1(2)(iii)) and W is doubly stochastic, these matrices commute with J and preserve the consensus/disagreement decomposition; in particular, $\widetilde{A\mathbf{Z}} = A\tilde{\mathbf{Z}}$, $\widetilde{C\mathbf{Z}} = C\tilde{\mathbf{Z}}$, and $\widetilde{B^2\mathbf{Z}} = B^2\tilde{\mathbf{Z}}$. Applying $(I - J)$ to (4.34) yields

$$\tilde{\mathbf{X}}^{k+1} = (AC - B^2)\tilde{\mathbf{X}}^k - \alpha A\tilde{\mathbf{U}}^{k+1} - \tilde{\mathbf{s}}^k, \quad (4.35a)$$

$$\tilde{\mathbf{s}}^{k+1} = \tilde{\mathbf{s}}^k + B^2\tilde{\mathbf{X}}^k. \quad (4.35b)$$

Step 3: diagonalization in the eigenbasis of W . Since W is symmetric, it admits an orthogonal eigendecomposition $W = U \text{diag}(1, \hat{\Lambda}) U^\top$ with $U = [\frac{1}{\sqrt{N}}\mathbf{1}, \hat{U}]$ and $\hat{U}\hat{U}^\top = I - J$. Because A, C, B^2 are polynomials in W , they share the eigenvectors of W and thus can be decomposed as

$$A = U \text{diag}(1, \hat{\Lambda}_a) U^\top, \quad C = U \text{diag}(1, \hat{\Lambda}_c) U^\top, \quad B^2 = U \text{diag}(0, \hat{\Lambda}_b^2) U^\top,$$

for some diagonal $\hat{\Lambda}_a, \hat{\Lambda}_c, \hat{\Lambda}_b^2 \in \mathbb{R}^{(N-1) \times (N-1)}$. Define the projected variables

$$\hat{\mathbf{X}}^k \triangleq \hat{\mathbf{U}}^\top \mathbf{X}^k, \quad \hat{\mathbf{s}}^k \triangleq \hat{\mathbf{U}}^\top \mathbf{s}^k, \quad \hat{\mathbf{U}}^{k+1} \triangleq \hat{\mathbf{U}}^\top \tilde{\mathbf{U}}^{k+1}.$$

Then (4.35) is equivalent to

$$\hat{\mathbf{X}}^{k+1} = (\hat{\Lambda}_a \hat{\Lambda}_c - \hat{\Lambda}_b^2) \hat{\mathbf{X}}^k - \hat{\mathbf{s}}^k - \alpha \hat{\Lambda}_a \hat{\mathbf{U}}^{k+1}, \quad (4.36a)$$

$$\hat{\mathbf{s}}^{k+1} = \hat{\mathbf{s}}^k + \hat{\Lambda}_b^2 \hat{\mathbf{X}}^k. \quad (4.36b)$$

Introduce the augmented state $\hat{\zeta}^k := [\hat{\mathbf{X}}^k; \hat{\mathbf{s}}^k]$ and the deviation matrix

$$\mathbf{G} \triangleq \begin{bmatrix} \hat{\Lambda}_a \hat{\Lambda}_c - \hat{\Lambda}_b^2 & -I \\ \hat{\Lambda}_b^2 & I \end{bmatrix}.$$

Then

$$\hat{\zeta}^{k+1} = \mathbf{G} \hat{\zeta}^k - \alpha \begin{bmatrix} \hat{\Lambda}_a \hat{\mathbf{U}}^{k+1} \\ 0 \end{bmatrix}. \quad (4.37)$$

By the stability assumption on $\mathbf{G}$, there exist an invertible matrix $\hat{\mathbf{V}}$ and a matrix Γ such that $\mathbf{G} = \hat{\mathbf{V}} \Gamma \hat{\mathbf{V}}^{-1}$ and $\|\Gamma\| =: \gamma < 1$. Define $v_1 := \|\hat{\mathbf{V}}\|$, $v_2 := \|\hat{\mathbf{V}}^{-1}\|$, and $\lambda_a := \|\hat{\Lambda}_a\|$.

Step 4: contraction and unrolling. Let $\hat{e}^k := \hat{\mathbf{V}}^{-1} \hat{\zeta}^k$. Multiplying (4.37) by $\hat{\mathbf{V}}^{-1}$ gives

$$\hat{e}^{k+1} = \Gamma \hat{e}^k - \alpha \hat{\mathbf{V}}^{-1} \begin{bmatrix} \hat{\Lambda}_a \hat{\mathbf{U}}^{k+1} \\ 0 \end{bmatrix}.$$

Taking Frobenius norms and using $\|\hat{\mathbf{U}}^{k+1}\|_F \leq \|\tilde{\mathbf{U}}^{k+1}\|_F$ yields

$$\|\hat{e}^{k+1}\|_F \leq \gamma \|\hat{e}^k\|_F + \alpha v_2 \lambda_a \|\tilde{\mathbf{U}}^{k+1}\|_F.$$

Unrolling gives

$$\|\hat{e}^k\|_F \leq \gamma^k \|\hat{e}^0\|_F + \frac{\alpha v_2 \lambda_a}{1 - \gamma} \sup_{0 \leq t \leq k} \|\tilde{\mathbf{U}}^{t+1}\|_F.$$

Since $\hat{\zeta}^k = \hat{\mathbf{V}} \hat{e}^k$, we have $\|\hat{\mathbf{X}}^k\|_F \leq \|\hat{\zeta}^k\|_F \leq v_1 \|\hat{e}^k\|_F$ and $\|\hat{e}^0\|_F \leq v_2 \|\hat{\zeta}^0\|_F$. Moreover, $\mathbf{Y}^0 = 0$ implies $\mathbf{s}^0 = -B^2 \mathbf{X}^0$ and hence $\hat{\mathbf{s}}^0 = -\hat{\Lambda}_b^2 \hat{\mathbf{X}}^0$. Therefore $\|\hat{\zeta}^0\|_F \leq (1 + \|\hat{\Lambda}_b^2\|) \|\hat{\mathbf{X}}^0\|_F \leq (1 + \|B^2\|) \|\tilde{\mathbf{X}}^0\|_F$. Finally, since $\tilde{\mathbf{U}} \tilde{\mathbf{U}}^\top = I - J$ we have $\|\hat{\mathbf{X}}^k\|_F = \|\tilde{\mathbf{X}}^k\|_F$. Combining these bounds yields (4.31).

Step 5: boundedness of the Muon directions. If $\mathbf{U}^{k+1} = \text{msgn}(\mathbf{H}^{k+1})$, then by Lemma 4.1, each block satisfies $\|U_i^{k+1}\|_F \leq \sqrt{r_{\max}}$. Thus

$$\|\tilde{\mathbf{U}}^{k+1}\|_F \leq \|\mathbf{U}^{k+1}\|_F + \|J \mathbf{U}^{k+1}\|_F \leq 2 \|\mathbf{U}^{k+1}\|_F \leq 2 \sqrt{\sum_{i=1}^N \|U_i^{k+1}\|_F^2} \leq 2 \sqrt{N r_{\max}},$$

which is (4.32). □

Drift, Mismatch, and Tracking Error Decomposition

Lemma 4.8 (EMA bias for averaged true gradients (drift term)). *Suppose Assumption 2 holds. Define the averaged true gradient at the local iterates*

$$\bar{\nabla} \mathbf{f}(\mathbf{X}^k) \triangleq \frac{1}{N} \sum_{i=1}^N \nabla f_i(\mathbf{X}_i^k),$$

and the averaged EMA of true gradients by

$$\bar{C}^0 := \bar{\nabla} \mathbf{f}(\mathbf{X}^0), \quad \bar{C}^{k+1} := \beta \bar{C}^k + (1-\beta) \bar{\nabla} \mathbf{f}(\mathbf{X}^k), \quad k \geq 0.$$

Define the (one-step-ahead) EMA bias

$$D^k \triangleq \bar{\nabla} \mathbf{f}(\mathbf{X}^k) - \bar{C}^{k+1}, \quad k \geq 0.$$

Then, for all $k \geq 0$, D^k obeys the recursion

$$D^{k+1} = \beta D^k + \beta (\bar{\nabla} \mathbf{f}(\mathbf{X}^{k+1}) - \bar{\nabla} \mathbf{f}(\mathbf{X}^k)).$$

Moreover, for all $k \geq 0$,

$$\|\bar{\nabla} \mathbf{f}(\mathbf{X}^{k+1}) - \bar{\nabla} \mathbf{f}(\mathbf{X}^k)\|_* \leq \frac{L_*}{N} \sum_{i=1}^N \|\mathbf{X}_i^{k+1} - \mathbf{X}_i^k\|. \quad (4.38)$$

If, in addition, the iterates are generated by Algorithm 1, then

$$\frac{1}{N} \sum_{i=1}^N \|\mathbf{X}_i^{k+1} - \mathbf{X}_i^k\| \leq \|\bar{\mathbf{X}}^{k+1} - \bar{\mathbf{X}}^k\| + \frac{1}{\sqrt{N}} \|\tilde{\mathbf{X}}^{k+1}\|_F + \frac{1}{\sqrt{N}} \|\tilde{\mathbf{X}}^k\|_F, \quad (4.39)$$

where $\bar{\mathbf{X}}^k := \frac{1}{N} \sum_i \mathbf{X}_i^k$ and $\tilde{\mathbf{X}}^k := (I - J)\mathbf{X}^k$ with $J := \frac{1}{N} \mathbf{1}\mathbf{1}^\top$. Consequently, using Lemmas 4.2 and 4.3 (which imply $\|\bar{\mathbf{X}}^{k+1} - \bar{\mathbf{X}}^k\| \leq \alpha$), we have for all $k \geq 0$,

$$\|D^k\|_* \leq \frac{\beta L_*}{1-\beta} \left(\alpha + \frac{2}{\sqrt{N}} \sup_{t \geq 0} (\|\tilde{\mathbf{X}}^{t+1}\|_F + \|\tilde{\mathbf{X}}^t\|_F) \right). \quad (4.40)$$

In particular, if Lemma 4.7 holds (so that for all $t \geq 0$, $\|\tilde{\mathbf{X}}^t\|_F \leq v_1 v_2 \gamma^t \|\tilde{\mathbf{X}}^0\|_F + \frac{2\alpha v_1 v_2 \lambda_a}{1-\gamma} \sqrt{N r_{\max}}$), then for all $k \geq 0$,

$$\|D^k\|_* \leq \frac{\beta L_*}{1-\beta} \left[\alpha + \frac{4v_1 v_2}{\sqrt{N}} \|\tilde{\mathbf{X}}^0\|_F + \frac{8\alpha v_1 v_2 \lambda_a}{1-\gamma} \sqrt{r_{\max}} \right]. \quad (4.41)$$

Proof. Throughout, abbreviate $g^k := \bar{\nabla} \mathbf{f}(\mathbf{X}^k)$.

Step 1: bias recursion. By the definition of D^{k+1} and the EMA update,

$$D^{k+1} = g^{k+1} - \bar{C}^{k+2} = g^{k+1} - (\beta \bar{C}^{k+1} + (1-\beta)g^{k+1}) = \beta(g^{k+1} - \bar{C}^{k+1}).$$

Next, add and subtract g^k and use $D^k = g^k - \bar{C}^{k+1}$ to obtain

$$g^{k+1} - \bar{C}^{k+1} = (g^{k+1} - g^k) + (g^k - \bar{C}^{k+1}) = (g^{k+1} - g^k) + D^k.$$

Combining the last two displays yields

$$D^{k+1} = \beta D^k + \beta (g^{k+1} - g^k),$$

as claimed.

Step 2: bound the averaged true-gradient drift by smoothness. By convexity of $\|\cdot\|_*$ and Assumption 2,

$$\begin{aligned} \|g^{k+1} - g^k\|_* &= \left\| \frac{1}{N} \sum_{i=1}^N (\nabla f_i(X_i^{k+1}) - \nabla f_i(X_i^k)) \right\|_* \\ &\leq \frac{1}{N} \sum_{i=1}^N \|\nabla f_i(X_i^{k+1}) - \nabla f_i(X_i^k)\|_* \\ &\leq \frac{L_*}{N} \sum_{i=1}^N \|X_i^{k+1} - X_i^k\|, \end{aligned}$$

which is (4.38).

Step 3: bound the average step size by the average motion and consensus errors. For each i , insert and subtract the averages $\bar{X}^{k+1}$ and $\bar{X}^k$ and use triangle inequality:

$$\|X_i^{k+1} - X_i^k\| \leq \|X_i^{k+1} - \bar{X}^{k+1}\| + \|\bar{X}^{k+1} - \bar{X}^k\| + \|\bar{X}^k - X_i^k\|.$$

Averaging over i yields

$$\frac{1}{N} \sum_{i=1}^N \|X_i^{k+1} - X_i^k\| \leq \|\bar{X}^{k+1} - \bar{X}^k\| + \frac{1}{N} \sum_{i=1}^N \|X_i^{k+1} - \bar{X}^{k+1}\| + \frac{1}{N} \sum_{i=1}^N \|X_i^k - \bar{X}^k\|.$$

For the disagreement terms, use $\|A\| \leq \|A\|_F$ and Cauchy-Schwarz:

$$\frac{1}{N} \sum_{i=1}^N \|X_i^k - \bar{X}^k\| \leq \frac{1}{N} \sum_{i=1}^N \|X_i^k - \bar{X}^k\|_F \leq \frac{1}{\sqrt{N}} \left(\sum_{i=1}^N \|X_i^k - \bar{X}^k\|_F^2 \right)^{1/2} = \frac{1}{\sqrt{N}} \|\tilde{\mathbf{X}}^k\|_F \leq \frac{2}{\sqrt{N}} \|\tilde{\mathbf{X}}^k\|_F.$$

The same bound holds with k replaced by $k+1$, yielding (4.39).

Step 4: unroll the bias recursion. Combine the recursion from Step 1, the drift bound (4.38), and the step-size bound (4.39) to get

$$\|D^{k+1}\|_* \leq \beta \|D^k\|_* + \beta L_* \left(\|\bar{X}^{k+1} - \bar{X}^k\| + \frac{2}{\sqrt{N}} \|\tilde{\mathbf{X}}^{k+1}\|_F + \frac{2}{\sqrt{N}} \|\tilde{\mathbf{X}}^k\|_F \right).$$

Under Algorithm 1, Lemma 4.2 gives $\|\bar{X}^{k+1} - \bar{X}^k\| = \alpha \|\bar{S}^{k+1}\|$, and Lemma 4.3 (specifically (4.16) therein) implies $\|\bar{S}^{k+1}\| \leq 1$, hence $\|\bar{X}^{k+1} - \bar{X}^k\| \leq \alpha$. Moreover, $D^0 = g^0 - \bar{C}^1 = 0$ because $\bar{C}^0 = g^0$. Unrolling the above linear recursion and bounding the geometric series yields

$$\|D^k\|_* \leq \beta L_* \sum_{t=0}^{k-1} \beta^{k-1-t} \left(\alpha + \frac{2}{\sqrt{N}} \|\tilde{\mathbf{X}}^{t+1}\|_F + \frac{2}{\sqrt{N}} \|\tilde{\mathbf{X}}^t\|_F \right) \leq \frac{\beta L_*}{1-\beta} \left(\alpha + \frac{2}{\sqrt{N}} \sup_{t \geq 0} (\|\tilde{\mathbf{X}}^{t+1}\|_F + \|\tilde{\mathbf{X}}^t\|_F) \right),$$

which proves (4.40).

Step 5: plug in the SUDA consensus bound (Lemma 4.7). Lemma 4.7 implies for all $t \geq 0$,

$$\|\tilde{\mathbf{X}}^t\|_F \leq v_1 v_2 \|\tilde{\mathbf{X}}^0\|_F + \frac{2\alpha v_1 v_2 \lambda_a}{1-\gamma} \sqrt{Nr_{\max}},$$

since $\gamma^t \leq 1$. Therefore,

$$\sup_{t \geq 0} (\|\tilde{\mathbf{X}}^{t+1}\|_F + \|\tilde{\mathbf{X}}^t\|_F) \leq 2v_1 v_2 \|\tilde{\mathbf{X}}^0\|_F + \frac{4\alpha v_1 v_2 \lambda_a}{1-\gamma} \sqrt{Nr_{\max}}.$$

Substituting this into (4.40) and simplifying gives (4.41). $\square$

Lemma 4.9 (Consensus-induced gradient mismatch). *Assume Assumption 2. For any stacked primal iterate $\mathbf{X}^k = \text{col}\{X_1^k, \dots, X_N^k\}$, define*

$$\bar{X}^k := \frac{1}{N} \sum_{i=1}^N X_i^k, \quad \tilde{\mathbf{X}}^k := (I - J)\mathbf{X}^k, \quad J := \frac{1}{N} \mathbf{1}\mathbf{1}^\top.$$

Then the mismatch between the gradient at the averaged point and the average of local gradients at local points satisfies

$$\left\| \nabla f(\bar{X}^k) - \frac{1}{N} \sum_{i=1}^N \nabla f_i(X_i^k) \right\|_* \leq \frac{L_*}{\sqrt{N}} \|\tilde{\mathbf{X}}^k\|_F. \quad (4.42)$$

Proof. Using $\nabla f(\bar{X}^k) = \frac{1}{N} \sum_{i=1}^N \nabla f_i(\bar{X}^k)$, we write

$$\nabla f(\bar{X}^k) - \frac{1}{N} \sum_{i=1}^N \nabla f_i(X_i^k) = \frac{1}{N} \sum_{i=1}^N (\nabla f_i(\bar{X}^k) - \nabla f_i(X_i^k)).$$

By convexity of the nuclear norm and the triangle inequality,

$$\left\| \nabla f(\bar{X}^k) - \frac{1}{N} \sum_{i=1}^N \nabla f_i(X_i^k) \right\|_* \leq \frac{1}{N} \sum_{i=1}^N \|\nabla f_i(\bar{X}^k) - \nabla f_i(X_i^k)\|_*.$$

By Assumption 2, $\|\nabla f_i(\bar{X}^k) - \nabla f_i(X_i^k)\|_* \leq L_* \|\bar{X}^k - X_i^k\|$. Moreover, $\|\bar{X}^k - X_i^k\| \leq \|\bar{X}^k - X_i^k\|_F$. Therefore,

$$\left\| \nabla f(\bar{X}^k) - \frac{1}{N} \sum_{i=1}^N \nabla f_i(X_i^k) \right\|_* \leq \frac{L_*}{N} \sum_{i=1}^N \|X_i^k - \bar{X}^k\|_F.$$

Finally, Cauchy–Schwarz gives

$$\frac{1}{N} \sum_{i=1}^N \|X_i^k - \bar{X}^k\|_F \leq \frac{1}{\sqrt{N}} \left(\sum_{i=1}^N \|X_i^k - \bar{X}^k\|_F^2 \right)^{1/2} = \frac{1}{\sqrt{N}} \|\bar{\mathbf{X}}^k\|_F.$$

Combining the last two displays yields (4.42). $\square$

Lemma 4.10 (Decomposing the tracking error $\|\nabla f(\bar{X}^k) - H_i^{k+1}\|_*$). *Assume Assumptions 1–3. Let $(\mathbf{X}^k, \mathbf{M}^k, \mathbf{H}^k)$ be generated by (3.1) with initialization $H_i^0 = M_i^0$ for all $i \in [N]$. Define*

$$\bar{X}^k := \frac{1}{N} \sum_{i=1}^N X_i^k, \quad \bar{M}^k := \frac{1}{N} \sum_{i=1}^N M_i^k, \quad \bar{H}^k := \frac{1}{N} \sum_{i=1}^N H_i^k,$$

$$\tilde{\mathbf{X}}^k := (I - J)\mathbf{X}^k, \quad \tilde{\mathbf{H}}^k := (I - J)\mathbf{H}^k, \quad J := \frac{1}{N} \mathbf{1}\mathbf{1}^\top.$$

Moreover, define the averaged true-gradient EMA $(\bar{C}^k)_{k \geq 0}$ and the drift variable $(D^k)_{k \geq 0}$ as in Lemma 4.8:

$$\bar{C}^0 := \bar{\nabla} \mathbf{f}(\mathbf{X}^0), \quad \bar{C}^{k+1} := \beta \bar{C}^k + (1 - \beta) \bar{\nabla} \mathbf{f}(\mathbf{X}^k), \quad D^k := \bar{\nabla} \mathbf{f}(\mathbf{X}^k) - \bar{C}^{k+1},$$

where $\bar{\nabla} \mathbf{f}(\mathbf{X}^k) := \frac{1}{N} \sum_{i=1}^N \nabla f_i(X_i^k)$. Then for every $k \geq 0$,

$$\begin{aligned} \frac{1}{N} \sum_{i=1}^N \|\nabla f(\bar{X}^k) - H_i^{k+1}\|_* &\leq \frac{L_*}{\sqrt{N}} \|\tilde{\mathbf{X}}^k\|_F + \|D^k\|_* \\ &\quad + \|\bar{C}^{k+1} - \bar{M}^{k+1}\|_* + \frac{1}{N} \sum_{i=1}^N \|H_i^{k+1} - \bar{H}^{k+1}\|_*. \end{aligned} \tag{4.43}$$

Furthermore, letting $r_{\max} := \min\{m, n\}$, we have the conversion bound

$$\frac{1}{N} \sum_{i=1}^N \|H_i^{k+1} - \bar{H}^{k+1}\|_* \leq \sqrt{\frac{r_{\max}}{N}} \|\tilde{\mathbf{H}}^{k+1}\|_F. \tag{4.44}$$

Also,

$$\|\bar{C}^{k+1} - \bar{M}^{k+1}\|_* \leq \sqrt{r_{\max}} \|\bar{C}^{k+1} - \bar{M}^{k+1}\|_F. \tag{4.45}$$

Proof. Step 1: a four-term decomposition. Fix $k \geq 0$ and an agent $i \in [N]$. Using Lemma 4.5, we have $\bar{H}^{k+1} = \bar{M}^{k+1}$. Therefore, by repeated addition/subtraction and the triangle inequality,

$$\begin{aligned} \|\nabla f(\bar{X}^k) - H_i^{k+1}\|_* &\leq \left\| \nabla f(\bar{X}^k) - \bar{\nabla} \mathbf{f}(\mathbf{X}^k) \right\|_* + \left\| \bar{\nabla} \mathbf{f}(\mathbf{X}^k) - \bar{C}^{k+1} \right\|_* \\ &\quad + \|\bar{C}^{k+1} - \bar{M}^{k+1}\|_* + \|\bar{H}^{k+1} - H_i^{k+1}\|_* \\ &= \left\| \nabla f(\bar{X}^k) - \bar{\nabla} \mathbf{f}(\mathbf{X}^k) \right\|_* + \|D^k\|_* + \|\bar{C}^{k+1} - \bar{M}^{k+1}\|_* + \|H_i^{k+1} - \bar{H}^{k+1}\|_*. \end{aligned}$$

Averaging over i yields

$$\begin{aligned} \frac{1}{N} \sum_{i=1}^N \|\nabla f(\bar{X}^k) - H_i^{k+1}\|_* &\leq \left\| \nabla f(\bar{X}^k) - \bar{\nabla} \mathbf{f}(\mathbf{X}^k) \right\|_* + \|D^k\|_* \\ &\quad + \|\bar{C}^{k+1} - \bar{M}^{k+1}\|_* + \frac{1}{N} \sum_{i=1}^N \|H_i^{k+1} - \bar{H}^{k+1}\|_*. \end{aligned}$$

Step 2: bound the first term by disagreement of $\mathbf{X}^k$. Apply Lemma 4.9 to obtain

$$\left\| \nabla f(\bar{X}^k) - \bar{\nabla} \mathbf{f}(\mathbf{X}^k) \right\|_* \leq \frac{L_*}{\sqrt{N}} \|\bar{\mathbf{X}}^k\|_F,$$

which proves (4.43).

Step 3: disagreement-to-average conversion for $\mathbf{H}^{k+1}$. For each i , by Cauchy–Schwarz on singular values, $\|A\|_* \leq \sqrt{r_{\max}} \|A\|_F$. Therefore,

$$\frac{1}{N} \sum_{i=1}^N \|H_i^{k+1} - \bar{H}^{k+1}\|_* \leq \frac{\sqrt{r_{\max}}}{N} \sum_{i=1}^N \|H_i^{k+1} - \bar{H}^{k+1}\|_F \leq \sqrt{\frac{r_{\max}}{N}} \left(\sum_{i=1}^N \|H_i^{k+1} - \bar{H}^{k+1}\|_F^2 \right)^{1/2}.$$

Since $\tilde{\mathbf{H}}^{k+1} = (I - J)\mathbf{H}^{k+1}$ has i -th block $H_i^{k+1} - \bar{H}^{k+1}$, $\|\tilde{\mathbf{H}}^{k+1}\|_F^2 = \sum_{i=1}^N \|H_i^{k+1} - \bar{H}^{k+1}\|_F^2$, proving (4.44).

Step 4: nuclear-to-Frobenius conversion for the averaged EMA noise. The same inequality $\|A\|_* \leq \sqrt{r_{\max}} \|A\|_F$ yields (4.45). $\square$

Forcing Terms and Disagreement Bounds

Lemma 4.11 (A uniform bound on the stacked one-step motion). *Assume Assumptions 1–4 and that the iterates are generated by (3.1) with $\mathbf{Y}_i^0 = 0$. Define the agentwise average $\bar{\mathbf{X}}^k := \frac{1}{N} \sum_{i=1}^N \mathbf{X}_i^k$, let $J := \frac{1}{N} \mathbf{1}\mathbf{1}^\top$, and define $\tilde{\mathbf{X}}^k := (I - J)\mathbf{X}^k$. Then, for every $k \geq 0$,*

$$\|\mathbf{X}^{k+1} - \mathbf{X}^k\|_F \leq \sqrt{N} \|\bar{\mathbf{X}}^{k+1} - \bar{\mathbf{X}}^k\|_F + \|\tilde{\mathbf{X}}^{k+1}\|_F + \|\tilde{\mathbf{X}}^k\|_F. \quad (4.46)$$

Consequently, using Lemma 4.2 and Lemma 4.1 (which imply $\|\bar{X}^{k+1} - \bar{X}^k\|_F \leq \alpha\sqrt{r_{\max}}$), we have

$$\|\mathbf{X}^{k+1} - \mathbf{X}^k\|_F \leq \alpha\sqrt{Nr_{\max}} + \|\tilde{\mathbf{X}}^{k+1}\|_F + \|\tilde{\mathbf{X}}^k\|_F, \quad r_{\max} := \min\{m, n\}. \quad (4.47)$$

If, moreover, the assumptions and initialization of Lemma 4.7 hold with $\mathbf{U}^{k+1} = \text{msgn}(\mathbf{H}^{k+1})$ (so that $\|\tilde{\mathbf{X}}^t\|_F \leq v_1 v_2 \gamma^t \|\tilde{\mathbf{X}}^0\|_F + \frac{2\alpha v_1 v_2 \lambda_a}{1-\gamma} \sqrt{Nr_{\max}}$ for all t), then

$$\sup_{k \geq 0} \|\mathbf{X}^{k+1} - \mathbf{X}^k\|_F \leq \alpha\sqrt{Nr_{\max}} + 2v_1 v_2 \|\tilde{\mathbf{X}}^0\|_F + \frac{4\alpha v_1 v_2 \lambda_a}{1-\gamma} \sqrt{Nr_{\max}}. \quad (4.48)$$

Proof. Step 1: decompose into average and disagreement parts. Write

$$\mathbf{X}^{k+1} - \mathbf{X}^k = J(\mathbf{X}^{k+1} - \mathbf{X}^k) + (I - J)(\mathbf{X}^{k+1} - \mathbf{X}^k).$$

Take Frobenius norms and use the triangle inequality:

$$\|\mathbf{X}^{k+1} - \mathbf{X}^k\|_F \leq \|J(\mathbf{X}^{k+1} - \mathbf{X}^k)\|_F + \|(I - J)(\mathbf{X}^{k+1} - \mathbf{X}^k)\|_F.$$

Because $J\mathbf{Z}$ has all blocks equal to $\bar{\mathbf{Z}} := \frac{1}{N} \sum_i \mathbf{Z}_i$, we have $J(\mathbf{X}^{k+1} - \mathbf{X}^k) = \text{col}\{\bar{X}^{k+1} - \bar{X}^k, \dots, \bar{X}^{k+1} - \bar{X}^k\}$ and hence

$$\|J(\mathbf{X}^{k+1} - \mathbf{X}^k)\|_F = \sqrt{N} \|\bar{X}^{k+1} - \bar{X}^k\|_F.$$

Moreover, $(I - J)(\mathbf{X}^{k+1} - \mathbf{X}^k) = \tilde{\mathbf{X}}^{k+1} - \tilde{\mathbf{X}}^k$, so

$$\|(I - J)(\mathbf{X}^{k+1} - \mathbf{X}^k)\|_F \leq \|\tilde{\mathbf{X}}^{k+1}\|_F + \|\tilde{\mathbf{X}}^k\|_F.$$

Combining the last three displays yields (4.46).

Step 2: bound the average motion in Frobenius norm. By Lemma 4.2, $\bar{X}^{k+1} - \bar{X}^k = -\alpha \bar{S}^{k+1}$, where $\bar{S}^{k+1} = \frac{1}{N} \sum_{i=1}^N S_i^{k+1}$ and $S_i^{k+1} = \text{msgn}(H_i^{k+1})$. By Lemma 4.1(2), $\|S_i^{k+1}\|_F \leq \sqrt{r_{\max}}$. By convexity of the Frobenius norm, $\|\bar{S}^{k+1}\|_F \leq \frac{1}{N} \sum_i \|S_i^{k+1}\|_F \leq \sqrt{r_{\max}}$. Hence $\|\bar{X}^{k+1} - \bar{X}^k\|_F \leq \alpha\sqrt{r_{\max}}$. Substituting this into (4.46) yields (4.47).

Step 3: plug in the SUDA consensus bound. Under Lemma 4.7 with Muon directions, for any $t \geq 0$, $\|\tilde{\mathbf{X}}^t\|_F \leq v_1 v_2 \|\tilde{\mathbf{X}}^0\|_F + \frac{2\alpha v_1 v_2 \lambda_a}{1-\gamma} \sqrt{Nr_{\max}}$ since $\gamma^t \leq 1$. Plugging this into (4.47) and taking the supremum over k yields (4.48). $\square$

Lemma 4.12 (Bounding the gradient-difference forcing term). *Assume Assumptions 2 and 3. Let $\mathbf{X}^k$ be any (possibly random) stacked iterates and let $\mathbf{G}^k = \text{col}\{G_1^k, \dots, G_N^k\}$ be the stochastic gradients with $G_i^k = \nabla F_i(X_i^k; \zeta_i^k)$. Define the stacked noise $\Delta_i^k := G_i^k - \nabla f_i(X_i^k)$ and $\Delta^k := \text{col}\{\Delta_1^k, \dots, \Delta_N^k\}$. Let $J := \frac{1}{N} \mathbf{1}\mathbf{1}^\top$ and $\tilde{\mathbf{Z}}^k := (I - J)\mathbf{Z}^k$. Then for any $k \geq 1$,*

$$\mathbb{E}[\|\tilde{\mathbf{G}}^k - \tilde{\mathbf{G}}^{k-1}\|_F^2] \leq 2L_*^2 \mathbb{E}[\|\mathbf{X}^k - \mathbf{X}^{k-1}\|_F^2] + 8N\sigma^2. \quad (4.49)$$

In particular, if Lemma 4.11 holds so that $\|\mathbf{X}^t - \mathbf{X}^{t-1}\|_F \leq B_X$ for all $t \geq 1$, then

$$\sup_{k \geq 1} \mathbb{E}[\|\tilde{\mathbf{G}}^k - \tilde{\mathbf{G}}^{k-1}\|_F^2] \leq 2L_*^2 B_X^2 + 8N\sigma^2. \quad (4.50)$$

Proof. Fix $k \geq 1$. Decompose

$$\mathbf{G}^k - \mathbf{G}^{k-1} = (\nabla \mathbf{f}(\mathbf{X}^k) - \nabla \mathbf{f}(\mathbf{X}^{k-1})) + (\Delta^k - \Delta^{k-1}),$$

where $\nabla \mathbf{f}(\mathbf{X}^k) := \text{col}\{\nabla f_1(X_1^k), \dots, \nabla f_N(X_N^k)\}$. Since $I - J$ is an orthogonal projection, $\|\tilde{\mathbf{Z}}\|_F \leq \|\mathbf{Z}\|_F$ for all stacked $\mathbf{Z}$. Therefore, using $\|U + V\|_F^2 \leq 2\|U\|_F^2 + 2\|V\|_F^2$,

$$\begin{aligned} \|\tilde{\mathbf{G}}^k - \tilde{\mathbf{G}}^{k-1}\|_F^2 &= \|(I - J)(\mathbf{G}^k - \mathbf{G}^{k-1})\|_F^2 \leq \|\mathbf{G}^k - \mathbf{G}^{k-1}\|_F^2 \\ &\leq 2\|\nabla \mathbf{f}(\mathbf{X}^k) - \nabla \mathbf{f}(\mathbf{X}^{k-1})\|_F^2 + 2\|\Delta^k - \Delta^{k-1}\|_F^2. \end{aligned}$$

Take expectations.

For the first term, by Assumption 2, for each i we have

$$\|\nabla f_i(X_i^k) - \nabla f_i(X_i^{k-1})\|_F \leq \|\nabla f_i(X_i^k) - \nabla f_i(X_i^{k-1})\|_* \leq L_* \|X_i^k - X_i^{k-1}\| \leq L_* \|X_i^k - X_i^{k-1}\|_F.$$

Squaring and summing over i yields $\|\nabla \mathbf{f}(\mathbf{X}^k) - \nabla \mathbf{f}(\mathbf{X}^{k-1})\|_F^2 \leq L_*^2 \|\mathbf{X}^k - \mathbf{X}^{k-1}\|_F^2$.

For the second term, use $\|A - B\|_F^2 \leq 2\|A\|_F^2 + 2\|B\|_F^2$ blockwise:

$$\|\Delta^k - \Delta^{k-1}\|_F^2 = \sum_{i=1}^N \|\Delta_i^k - \Delta_i^{k-1}\|_F^2 \leq 2 \sum_{i=1}^N \|\Delta_i^k\|_F^2 + 2 \sum_{i=1}^N \|\Delta_i^{k-1}\|_F^2.$$

Taking expectations and using Assumption 3 (unconditional version of the conditional variance bound) gives $\mathbb{E}\|\Delta_i^t\|_F^2 \leq \sigma^2$ for all i, t , and hence $\mathbb{E}\|\Delta^k - \Delta^{k-1}\|_F^2 \leq 4N\sigma^2$.

Combining the above bounds proves (4.49). Finally, if $\|\mathbf{X}^k - \mathbf{X}^{k-1}\|_F \leq B_X$ holds uniformly, then (4.50) follows immediately. $\square$

Lemma 4.13 (A bound on the EMA increment disagreement). *Assume Assumption 3. Let the iterates be generated by (3.1) with the EMA initialization $\mathbf{M}^0 = \mathbf{G}^0$ (as in Algorithm 1). Let $\beta \in [0, 1)$. Define $J := \frac{1}{N} \mathbf{1}\mathbf{1}^\top$ and $\tilde{\mathbf{Z}}^k := (I - J)\mathbf{Z}^k$. For $k \geq 1$ define the EMA increment $\mathbf{V}^k := \mathbf{M}^k - \mathbf{M}^{k-1}$ and its disagreement $\tilde{\mathbf{V}}^k := (I - J)\mathbf{V}^k$. Then $\mathbf{V}^1 = 0$ and for all $k \geq 1$,*

$$\tilde{\mathbf{V}}^{k+1} = \beta \tilde{\mathbf{V}}^k + (1 - \beta)(\tilde{\mathbf{G}}^k - \tilde{\mathbf{G}}^{k-1}), \quad \tilde{\mathbf{G}}^k := (I - J)\mathbf{G}^k. \quad (4.51)$$

Consequently, for any $\eta > 0$ and all $k \geq 1$,

$$\|\tilde{\mathbf{V}}^{k+1}\|_F^2 \leq (1 + \eta)\beta^2 \|\tilde{\mathbf{V}}^k\|_F^2 + (1 + 1/\eta)(1 - \beta)^2 \|\tilde{\mathbf{G}}^k - \tilde{\mathbf{G}}^{k-1}\|_F^2. \quad (4.52)$$

In particular, if there exists a constant B_G such that $\sup_{k \geq 1} \mathbb{E}\|\tilde{\mathbf{G}}^k - \tilde{\mathbf{G}}^{k-1}\|_F^2 \leq B_G^2$, then

$$\sup_{k \geq 1} \mathbb{E}\|\tilde{\mathbf{M}}^k - \tilde{\mathbf{M}}^{k-1}\|_F^2 = \sup_{k \geq 1} \mathbb{E}\|\tilde{\mathbf{V}}^k\|_F^2 \leq \frac{2(1 + \beta^2)}{(1 + \beta)^2} B_G^2 \leq 2B_G^2. \quad (4.53)$$

Proof. **Step 1: derive the increment recursion.** For $k \geq 1$, subtract the EMA updates at times k and $k-1$:

$$\mathbf{M}^{k+1} - \mathbf{M}^k = \beta(\mathbf{M}^k - \mathbf{M}^{k-1}) + (1-\beta)(\mathbf{G}^k - \mathbf{G}^{k-1}).$$

This is exactly $\mathbf{V}^{k+1} = \beta\mathbf{V}^k + (1-\beta)(\mathbf{G}^k - \mathbf{G}^{k-1})$. Applying $(I-J)$ to both sides gives (4.51). Moreover, since $\mathbf{M}^0 = \mathbf{G}^0$, we have $\mathbf{M}^1 = \beta\mathbf{M}^0 + (1-\beta)\mathbf{G}^0 = \mathbf{M}^0$ and hence $\mathbf{V}^1 = \mathbf{M}^1 - \mathbf{M}^0 = 0$.

Step 2: square and apply Young's inequality. From (4.51) and $\|U+V\|_F^2 \leq (1+\eta)\|U\|_F^2 + (1+1/\eta)\|V\|_F^2$, with $U = \beta\tilde{\mathbf{V}}^k$ and $V = (1-\beta)(\tilde{\mathbf{G}}^k - \tilde{\mathbf{G}}^{k-1})$, we get (4.52).

Step 3: uniform bound under a uniform forcing bound. Assume $\sup_{k \geq 1} \mathbb{E}\|\tilde{\mathbf{G}}^k - \tilde{\mathbf{G}}^{k-1}\|_F^2 \leq B_G^2$. For $\beta \in (0,1)$, pick $\eta := \frac{1-\beta^2}{2\beta^2}$ so that $(1+\eta)\beta^2 = \frac{1+\beta^2}{2} =: q \in (0,1)$ and $1+1/\eta = \frac{1+\beta^2}{1-\beta^2}$. Taking expectations in (4.52) gives, for all $k \geq 1$,

$$\mathbb{E}\|\tilde{\mathbf{V}}^{k+1}\|_F^2 \leq q\mathbb{E}\|\tilde{\mathbf{V}}^k\|_F^2 + (1-\beta)^2 \frac{1+\beta^2}{1-\beta^2} B_G^2.$$

Unrolling this recursion and using $\tilde{\mathbf{V}}^1 = 0$ yields, for all $k \geq 1$,

$$\mathbb{E}\|\tilde{\mathbf{V}}^{k+1}\|_F^2 \leq (1-\beta)^2 \frac{1+\beta^2}{1-\beta^2} \sum_{t=0}^{k-1} q^t B_G^2 \leq (1-\beta)^2 \frac{1+\beta^2}{1-\beta^2} \frac{1}{1-q} B_G^2.$$

Since $1-q = \frac{1-\beta^2}{2}$, the last display becomes

$$\mathbb{E}\|\tilde{\mathbf{V}}^{k+1}\|_F^2 \leq \frac{2(1+\beta^2)(1-\beta)^2}{(1-\beta^2)^2} B_G^2 = \frac{2(1+\beta^2)}{(1+\beta)^2} B_G^2.$$

Taking the supremum over k gives (4.53) for $\beta \in (0,1)$. For $\beta=0$, $\mathbf{V}^1=0$ and (4.51) reduces to $\tilde{\mathbf{V}}^{k+1} = \tilde{\mathbf{G}}^k - \tilde{\mathbf{G}}^{k-1}$, hence $\sup_{k \geq 1} \mathbb{E}\|\tilde{\mathbf{V}}^k\|_F^2 \leq B_G^2 \leq 2B_G^2$. $\square$

Lemma 4.14 (A uniform bound on the gradient-tracking disagreement). *Assume Assumption 1(1) holds. Let $\mathbf{H}^{k+1} = W(\mathbf{H}^k + \mathbf{M}^{k+1} - \mathbf{M}^k)$ as in (3.1). Let $\lambda := \rho(W-J) \in (0,1)$ and define $\tilde{\mathbf{Z}}^k := (I-J)\mathbf{Z}^k$. Assume that there exists a constant B_M such that*

$$\sup_{k \geq 0} \mathbb{E}\|\tilde{\mathbf{M}}^{k+1} - \tilde{\mathbf{M}}^k\|_F^2 \leq B_M^2. \quad (4.54)$$

Then for all $k \geq 0$,

$$\mathbb{E}\|\tilde{\mathbf{H}}^{k+1}\|_F^2 \leq \left(\frac{1+\lambda^2}{2}\right)^{k+1} \mathbb{E}\|\tilde{\mathbf{H}}^0\|_F^2 + \frac{2\lambda^2(1+\lambda^2)}{(1-\lambda^2)^2} B_M^2. \quad (4.55)$$

In particular,

$$\sup_{k \geq 0} \mathbb{E}\|\tilde{\mathbf{H}}^k\|_F^2 \leq \mathbb{E}\|\tilde{\mathbf{H}}^0\|_F^2 + \frac{2\lambda^2(1+\lambda^2)}{(1-\lambda^2)^2} B_M^2. \quad (4.56)$$

Proof. From Lemma 4.6, for any $\eta > 0$,

$$\|\tilde{\mathbf{H}}^{k+1}\|_F^2 \leq (1+\eta)\lambda^2 \|\tilde{\mathbf{H}}^k\|_F^2 + (1+1/\eta)\lambda^2 \|\tilde{\mathbf{M}}^{k+1} - \tilde{\mathbf{M}}^k\|_F^2.$$

Choose $\eta := \frac{1-\lambda^2}{2\lambda^2}$ so that $(1+\eta)\lambda^2 = \frac{1+\lambda^2}{2} =: q \in (0,1)$ and $1+1/\eta = 1 + \frac{2\lambda^2}{1-\lambda^2} = \frac{1+\lambda^2}{1-\lambda^2}$. Taking expectations and using (4.54) yields

$$\mathbb{E}\|\tilde{\mathbf{H}}^{k+1}\|_F^2 \leq q\mathbb{E}\|\tilde{\mathbf{H}}^k\|_F^2 + \lambda^2 \frac{1+\lambda^2}{1-\lambda^2} B_M^2.$$

Unrolling this linear recursion gives

$$\mathbb{E}\|\tilde{\mathbf{H}}^{k+1}\|_F^2 \leq q^{k+1}\mathbb{E}\|\tilde{\mathbf{H}}^0\|_F^2 + \lambda^2 \frac{1+\lambda^2}{1-\lambda^2} \sum_{t=0}^k q^t B_M^2.$$

Since $\sum_{t=0}^k q^t \leq \frac{1}{1-q} = \frac{2}{1-\lambda^2}$, we get

$$\mathbb{E}\|\tilde{\mathbf{H}}^{k+1}\|_F^2 \leq q^{k+1}\mathbb{E}\|\tilde{\mathbf{H}}^0\|_F^2 + \lambda^2 \frac{1+\lambda^2}{1-\lambda^2} \frac{2}{1-\lambda^2} B_M^2 = q^{k+1}\mathbb{E}\|\tilde{\mathbf{H}}^0\|_F^2 + \frac{2\lambda^2(1+\lambda^2)}{(1-\lambda^2)^2} B_M^2,$$

which is (4.55). Taking the supremum over k yields (4.56). $\square$

Lemma 4.15 (An explicit bound on the EMA increment disagreement). *Assume Assumptions 2 and 3. Let the iterates be generated by (3.1) with EMA parameter $\beta \in [0,1)$ and initialization $\mathbf{M}^0 = \mathbf{G}^0$ (as in Algorithm 1). Assume further that Lemma 4.11 holds so that there exists $B_X > 0$ with*

$$\|\mathbf{X}^k - \mathbf{X}^{k-1}\|_F \leq B_X, \quad \forall k \geq 1. \quad (4.57)$$

Define

$$B_G^2 := 2L_*^2 B_X^2 + 8N\sigma^2. \quad (4.58)$$

Then the forcing term in Lemma 4.6 admits the uniform bound

$$\sup_{k \geq 0} \mathbb{E}\|\tilde{\mathbf{M}}^{k+1} - \tilde{\mathbf{M}}^k\|_F^2 \leq 2B_G^2. \quad (4.59)$$

Proof. Lemma 4.12 and (4.57) imply $\sup_{k \geq 1} \mathbb{E}\|\tilde{\mathbf{G}}^k - \tilde{\mathbf{G}}^{k-1}\|_F^2 \leq B_G^2$. Then Lemma 4.13 yields $\sup_{k \geq 1} \mathbb{E}\|\tilde{\mathbf{M}}^k - \tilde{\mathbf{M}}^{k-1}\|_F^2 \leq 2B_G^2$. Renaming the index ($k \mapsto k+1$) gives (4.59). $\square$

Proofs of the Main Convergence Results

Proposition 4.3 (A closed stationarity bound). *Assume Assumptions 1–4. Let the iterates be generated by (3.1) with initialization $H_i^0 = M_i^0 = G_i^0$ and $Y_i^0 = 0$ for all $i \in [N]$. Assume additionally that $\mathbf{Y}^0 = 0$ and the stability condition in Lemma 4.7 hold (so that the constants $\gamma \in (0,1)$, $v_1, v_2 \geq 1$,*

and $\lambda_a \geq 0$ in Lemma 4.7 exist). Let $K \geq 1$. Define $\Delta_0 := \mathbb{E}[f(\bar{X}^0)] - f_{\inf}$ and $r_{\max} := \min\{m, n\}$. Then

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E}[\|\nabla f(\bar{X}^k)\|_*] \leq \frac{\Delta_0}{\alpha K} + \frac{L_* \alpha}{2} + 2\mathcal{T}_{\text{cons}}(K) + 2\mathcal{T}_{\text{drift}} + 2\mathcal{T}_{\text{noise}}(K) + 2\mathcal{T}_{\text{gt}}, \quad (4.60)$$

where

$$\mathcal{T}_{\text{cons}}(K) := \frac{L_*}{K\sqrt{N}} \sum_{k=0}^{K-1} \mathbb{E}\|\tilde{\mathbf{X}}^k\|_F \leq \frac{L_* v_1 v_2}{K\sqrt{N}} \frac{\|\tilde{\mathbf{X}}^0\|_F}{1-\gamma} + \frac{2L_* \alpha v_1 v_2 \lambda_a}{1-\gamma} \sqrt{r_{\max}}, \quad (4.61)$$

$$\mathcal{T}_{\text{drift}} := \sup_{k \geq 0} \mathbb{E}\|D^k\|_* \leq \frac{\beta L_*}{1-\beta} \left[\alpha + \frac{4v_1 v_2}{\sqrt{N}} \|\tilde{\mathbf{X}}^0\|_F + \frac{8\alpha v_1 v_2 \lambda_a}{1-\gamma} \sqrt{r_{\max}} \right], \quad (4.62)$$

$$\begin{aligned} \mathcal{T}_{\text{noise}}(K) &:= \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E}\|\bar{C}^{k+1} - \bar{M}^{k+1}\|_* \\ &\leq \sqrt{r_{\max}} \left[\sqrt{\frac{1-\beta}{1+\beta}} \frac{\sigma}{\sqrt{N}} + \frac{\sigma}{K\sqrt{N}} \frac{1}{1-\beta} \right], \end{aligned} \quad (4.63)$$

$$\mathcal{T}_{\text{gt}} := \sqrt{\frac{r_{\max}}{N}} \sup_{k \geq 0} \mathbb{E}\|\tilde{\mathbf{H}}^{k+1}\|_F \leq \sqrt{\frac{r_{\max}}{N}} \sqrt{\mathbb{E}\|\tilde{\mathbf{H}}^0\|_F^2 + \frac{4\lambda^2(1+\lambda^2)}{(1-\lambda^2)^2} B_G^2}, \quad (4.64)$$

with $\lambda := \rho(W - J) \in (0, 1)$ and

$$B_G^2 := 2L_*^2 B_X^2 + 8N\sigma^2, \quad B_X := \alpha \sqrt{Nr_{\max}} + 2v_1 v_2 \|\tilde{\mathbf{X}}^0\|_F + \frac{4\alpha v_1 v_2 \lambda_a}{1-\gamma} \sqrt{Nr_{\max}}. \quad (4.65)$$

Proof. Start from Proposition 4.2:

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E}\|\nabla f(\bar{X}^k)\|_* \leq \frac{\Delta_0}{\alpha K} + \frac{L_* \alpha}{2} + \frac{2}{K} \sum_{k=0}^{K-1} \mathbb{E} \left[\frac{1}{N} \sum_{i=1}^N \|\nabla f(\bar{X}^k) - H_i^{k+1}\|_* \right].$$

Apply Lemma 4.10 to the averaged tracking error inside the expectation and sum over k . This yields the decomposition into four contributions.

For the consensus term, apply Lemma 4.7 (with Muon directions) and sum the geometric series to obtain (4.61).

For the drift term, use the uniform bound from Lemma 4.8, giving (4.62).

For the stochastic EMA term, use Lemma 4.4 and the nuclear-to-Frobenius conversion (4.45) from Lemma 4.10; averaging β^{k+1} over $k=0, \dots, K-1$ yields (4.63).

For the gradient-tracking disagreement term, use (4.44) from Lemma 4.10 and Jensen/Cauchy-Schwarz: $\mathbb{E}\|\tilde{\mathbf{H}}^{k+1}\|_F \leq \sqrt{\mathbb{E}\|\tilde{\mathbf{H}}^{k+1}\|_F^2}$. Then apply Lemma 4.14 together with Lemma 4.15. Finally, invoke Lemma 4.11 to bound B_X as stated in (4.65). Collecting terms yields (4.60). $\square$

Lemma 4.16 (Averaged tracking error bound with $\sqrt{1-\beta}$ noise scaling). *Assume Assumptions 1–3. Let the iterates be generated by (3.1) with initialization $H_i^0 = M_i^0$ for all $i \in [N]$. Define the network averages*

$$\bar{X}^k := \frac{1}{N} \sum_{i=1}^N X_i^k, \quad \bar{H}^k := \frac{1}{N} \sum_{i=1}^N H_i^k, \quad \bar{M}^k := \frac{1}{N} \sum_{i=1}^N M_i^k.$$

Define the (average) tracking error

$$E_{\text{gt}}^k := \bar{H}^k - \nabla f(\bar{X}^k) \in \mathbb{R}^{m \times n}.$$

Let

$$\bar{G}^k := \frac{1}{N} \sum_{i=1}^N G_i^k, \quad \bar{g}^k := \frac{1}{N} \sum_{i=1}^N \nabla f_i(X_i^k), \quad \bar{\Delta}^k := \bar{G}^k - \bar{g}^k.$$

Then for every integer $K \geq 1$ we have

$$\begin{aligned} \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|E_{\text{gt}}^{k+1}\|_F &\leq \underbrace{\sqrt{\frac{1-\beta}{1+\beta}} \frac{\sigma}{\sqrt{N}}}_{\text{stochastic noise} \sim \frac{\sigma}{\sqrt{N}} \sqrt{1-\beta}} + \underbrace{\frac{L_*}{1-\beta} \cdot \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\bar{X}^{k+1} - \bar{X}^k\|_F}_{\text{drift} \sim \frac{L_* \alpha}{1-\beta}} \\ &\quad + \underbrace{\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\bar{g}^k - \nabla f(\bar{X}^k)\|_F}_{\text{consensus bias}} + \frac{1}{K} \sum_{k=0}^{K-1} \beta^{k+1} \mathbb{E} \|E_{\text{gt}}^0\|_F. \end{aligned} \quad (4.66)$$

Moreover, under Assumption 2, the consensus bias can be bounded by

$$\|\bar{g}^k - \nabla f(\bar{X}^k)\|_F \leq \frac{L_*}{\sqrt{N}} \|\tilde{\mathbf{X}}^k\|_F, \quad \tilde{\mathbf{X}}^k := (I - J) \mathbf{X}^k, \quad J := \frac{1}{N} \mathbf{1} \mathbf{1}^\top. \quad (4.67)$$

Proof. By Lemma 4.5, $\bar{H}^k = \bar{M}^k$ for all $k \geq 0$. Averaging the EMA update gives

$$\bar{M}^{k+1} = \beta \bar{M}^k + (1-\beta) \bar{G}^k. \quad (4.68)$$

Fix $k \geq 0$. Using $E_{\text{gt}}^k = \bar{M}^k - \nabla f(\bar{X}^k)$ and (4.68),

$$\begin{aligned} E_{\text{gt}}^{k+1} &= \bar{M}^{k+1} - \nabla f(\bar{X}^{k+1}) = \beta \bar{M}^k + (1-\beta) \bar{G}^k - \nabla f(\bar{X}^{k+1}) \\ &= \beta (\bar{M}^k - \nabla f(\bar{X}^k)) + (\nabla f(\bar{X}^k) - \nabla f(\bar{X}^{k+1})) + (1-\beta) (\bar{G}^k - \nabla f(\bar{X}^k)) \\ &= \beta E_{\text{gt}}^k + \underbrace{(1-\beta) \bar{\Delta}^k}_{\text{noise}} + \underbrace{(1-\beta) (\bar{g}^k - \nabla f(\bar{X}^k))}_{\text{consensus bias}} + \underbrace{(\nabla f(\bar{X}^k) - \nabla f(\bar{X}^{k+1}))}_{\text{drift}}. \end{aligned} \quad (4.69)$$

Step 1: decompose $E_{\text{gt}}^k = A^k + B^k$. Define $A^0 := 0$ and $B^0 := E_{\text{gt}}^0$, and for $k \geq 0$ define

$$A^{k+1} := \beta A^k + (1-\beta) (\bar{\Delta}^k + (\bar{g}^k - \nabla f(\bar{X}^k))), \quad B^{k+1} := \beta B^k + (\nabla f(\bar{X}^k) - \nabla f(\bar{X}^{k+1})).$$

Then (4.69) implies $E_{\text{gt}}^k = A^k + B^k$ for all k .

Step 2: bound the stochastic part (noise). Unrolling the A -recursion and keeping only the noise contribution yields

$$A_{\text{noise}}^{k+1} = (1-\beta) \sum_{t=0}^k \beta^{k-t} \bar{\Delta}^t.$$

By Assumption 3, conditional on $\mathcal{F}^t$ the noises are independent across agents and satisfy $\mathbb{E}[\bar{\Delta}^t | \mathcal{F}^t] = 0$ and $\mathbb{E}[\|\bar{\Delta}^t\|_F^2 | \mathcal{F}^t] \leq \sigma^2/N$. Therefore cross terms vanish, and

$$\mathbb{E}\|A_{\text{noise}}^{k+1}\|_F \leq \sqrt{\mathbb{E}\|A_{\text{noise}}^{k+1}\|_F^2} \leq (1-\beta) \sqrt{\sum_{t=0}^k \beta^{2(k-t)} \cdot \frac{\sigma^2}{N}} \leq (1-\beta) \sqrt{\frac{1}{1-\beta^2}} \cdot \frac{\sigma}{\sqrt{N}} = \sqrt{\frac{1-\beta}{1+\beta}} \frac{\sigma}{\sqrt{N}}.$$

Step 3: bound the drift part. Unrolling $B^{k+1} = \beta B^k + d^k$ with $d^k := \nabla f(\bar{X}^k) - \nabla f(\bar{X}^{k+1})$ gives

$$B^{k+1} = \beta^{k+1} B^0 + \sum_{t=0}^k \beta^{k-t} d^t.$$

Thus, by triangle inequality,

$$\mathbb{E}\|B^{k+1}\|_F \leq \beta^{k+1} \mathbb{E}\|B^0\|_F + \sum_{t=0}^k \beta^{k-t} \mathbb{E}\|d^t\|_F.$$

Now average over $k=0, \dots, K-1$ and exchange the order of summation:

$$\frac{1}{K} \sum_{k=0}^{K-1} \sum_{t=0}^k \beta^{k-t} \mathbb{E}\|d^t\|_F = \frac{1}{K} \sum_{t=0}^{K-1} \mathbb{E}\|d^t\|_F \sum_{k=t}^{K-1} \beta^{k-t} \leq \frac{1}{1-\beta} \cdot \frac{1}{K} \sum_{t=0}^{K-1} \mathbb{E}\|d^t\|_F.$$

Therefore,

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E}\|B^{k+1}\|_F \leq \frac{1}{K} \sum_{k=0}^{K-1} \beta^{k+1} \mathbb{E}\|B^0\|_F + \frac{1}{1-\beta} \cdot \frac{1}{K} \sum_{t=0}^{K-1} \mathbb{E}\|d^t\|_F.$$

Under Assumption 2, $\|d^t\|_F \leq \|d^t\|_* \leq L_* \|\bar{X}^{t+1} - \bar{X}^t\|$, and $\|\cdot\| \leq \|\cdot\|_F$. This yields the drift term in (4.66).

Step 4: bound the consensus bias. The consensus-bias contribution in A^{k+1} is

$$A_{\text{bias}}^{k+1} = (1-\beta) \sum_{t=0}^k \beta^{k-t} (\bar{g}^t - \nabla f(\bar{X}^t)).$$

By triangle inequality,

$$\mathbb{E}\|A_{\text{bias}}^{k+1}\|_F \leq (1-\beta) \sum_{t=0}^k \beta^{k-t} \mathbb{E}\|\bar{g}^t - \nabla f(\bar{X}^t)\|_F.$$

Now average over $k=0, \dots, K-1$ and exchange summations:

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|A_{\text{bias}}^{k+1}\|_F \leq \frac{1}{K} \sum_{t=0}^{K-1} \mathbb{E} \|\bar{g}^t - \nabla f(\bar{X}^t)\|_F \cdot (1-\beta) \sum_{k=t}^{K-1} \beta^{k-t} \leq \frac{1}{K} \sum_{t=0}^{K-1} \mathbb{E} \|\bar{g}^t - \nabla f(\bar{X}^t)\|_F,$$

where we used $(1-\beta) \sum_{s=0}^{\infty} \beta^s = 1$. This yields the bias term in (4.66).

Combining Steps 2–4 and using $E_{\text{gt}}^{k+1} = A^{k+1} + B^{k+1}$ proves (4.66). Finally, (4.67) follows from the same argument as Lemma 4.9:

$$\|\bar{g}^k - \nabla f(\bar{X}^k)\|_F = \left\| \frac{1}{N} \sum_{i=1}^N (\nabla f_i(X_i^k) - \nabla f_i(\bar{X}^k)) \right\|_F \leq \frac{1}{N} \sum_{i=1}^N L_* \|X_i^k - \bar{X}^k\|_F \leq \frac{L_*}{\sqrt{N}} \|\tilde{\mathbf{X}}^k\|_F.$$

□

Lemma 4.17 (Gradient-tracking disagreement bound (no heterogeneity term)). *Assume Assumptions 1(1), 2, and 3. Let $(\mathbf{X}^k, \mathbf{M}^k, \mathbf{H}^k)$ be generated by (3.1). Define the disagreement operator $\tilde{\mathbf{Z}} := (I - J)\mathbf{Z}$ with $J := \frac{1}{N}\mathbf{1}\mathbf{1}^\top$. Let $\lambda := \rho(W - J) \in (0, 1)$. Then, for every integer $K \geq 1$,*

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\tilde{\mathbf{H}}^{k+1}\|_F \leq \frac{\lambda}{K(1-\lambda)} \mathbb{E} \|\tilde{\mathbf{H}}^0\|_F + \frac{\lambda}{1-\lambda} \cdot \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\tilde{\mathbf{M}}^{k+1} - \tilde{\mathbf{M}}^k\|_F. \quad (4.70)$$

Moreover, with $r_{\max} := \min\{m, n\}$,

$$\begin{aligned} \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\tilde{\mathbf{M}}^{k+1} - \tilde{\mathbf{M}}^k\|_F &\leq \frac{\mathbb{E} \|\widetilde{\nabla \mathbf{f}}(\mathbf{X}^0) - \tilde{\mathbf{M}}^0\|_F}{K} + 2(1-\beta)\sigma\sqrt{N} \\ &\quad + L_* \left(\alpha \sqrt{Nr_{\max}} + \frac{2}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\tilde{\mathbf{X}}^k\|_F + \frac{1}{K} \mathbb{E} \|\tilde{\mathbf{X}}^K\|_F \right). \end{aligned} \quad (4.71)$$

Combining (4.70)–(4.71) yields

$$\begin{aligned} \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\tilde{\mathbf{H}}^{k+1}\|_F &\leq \frac{\lambda}{K(1-\lambda)} \mathbb{E} \|\tilde{\mathbf{H}}^0\|_F + \frac{\lambda}{1-\lambda} \left[\frac{\mathbb{E} \|\widetilde{\nabla \mathbf{f}}(\mathbf{X}^0) - \tilde{\mathbf{M}}^0\|_F}{K} \right. \\ &\quad \left. + L_* \left(\alpha \sqrt{Nr_{\max}} + \frac{2}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\tilde{\mathbf{X}}^k\|_F + \frac{1}{K} \mathbb{E} \|\tilde{\mathbf{X}}^K\|_F \right) + 2(1-\beta)\sigma\sqrt{N} \right]. \end{aligned} \quad (4.72)$$

In particular, if Lemma 4.7 holds with Muon directions, then

$$\begin{aligned} \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\tilde{\mathbf{H}}^{k+1}\|_F &\leq \frac{\lambda}{K(1-\lambda)} \mathbb{E} \|\tilde{\mathbf{H}}^0\|_F + \frac{\lambda}{1-\lambda} \left[\frac{\mathbb{E} \|\widetilde{\nabla \mathbf{f}}(\mathbf{X}^0) - \tilde{\mathbf{M}}^0\|_F}{K} + L_* \alpha \sqrt{Nr_{\max}} \right. \\ &\quad + \frac{2L_* v_1 v_2 (1 + \|B^2\|)}{K(1-\gamma)} \|\tilde{\mathbf{X}}^0\|_F + \frac{L_* v_1 v_2 (1 + \|B^2\|)}{K} \gamma^K \|\tilde{\mathbf{X}}^0\|_F + \frac{2L_* \alpha v_1 v_2 \lambda_a}{(1-\gamma)K} \sqrt{Nr_{\max}} \\ &\quad \left. + \frac{4L_* \alpha v_1 v_2 \lambda_a}{1-\gamma} \sqrt{Nr_{\max}} + 2(1-\beta)\sigma\sqrt{N} \right]. \end{aligned} \quad (4.73)$$

Proof. **Step 1: unroll the GT disagreement recursion.** From Lemma 4.6 (non-squared form),

$$\|\tilde{\mathbf{H}}^{k+1}\|_F \leq \lambda \|\tilde{\mathbf{H}}^k + (\tilde{\mathbf{M}}^{k+1} - \tilde{\mathbf{M}}^k)\|_F \leq \lambda \|\tilde{\mathbf{H}}^k\|_F + \lambda \|\tilde{\mathbf{M}}^{k+1} - \tilde{\mathbf{M}}^k\|_F.$$

Unrolling this scalar recursion and summing the resulting geometric series yields (4.70).

Step 2: bound the average EMA increment without a heterogeneity term. Define the local EMA tracking error

$$\mathbf{E}^k := \nabla \mathbf{f}(\mathbf{X}^k) - \mathbf{M}^k, \quad \tilde{\mathbf{E}}^k := (I - J)\mathbf{E}^k = \widetilde{\nabla \mathbf{f}}(\mathbf{X}^k) - \tilde{\mathbf{M}}^k.$$

From the EMA update $\mathbf{M}^{k+1} = \beta \mathbf{M}^k + (1 - \beta) \mathbf{G}^k$ and the decomposition $\mathbf{G}^k = \nabla \mathbf{f}(\mathbf{X}^k) + \Delta^k$, we have

$$\mathbf{E}^{k+1} = \nabla \mathbf{f}(\mathbf{X}^{k+1}) - \beta \mathbf{M}^k - (1 - \beta) \nabla \mathbf{f}(\mathbf{X}^k) - (1 - \beta) \Delta^k = \beta \mathbf{E}^k + (\nabla \mathbf{f}(\mathbf{X}^{k+1}) - \nabla \mathbf{f}(\mathbf{X}^k)) - (1 - \beta) \Delta^k.$$

Applying $(I - J)$ and using its non-expansiveness yields the recursion

$$\|\tilde{\mathbf{E}}^{k+1}\|_F \leq \beta \|\tilde{\mathbf{E}}^k\|_F + \|\widetilde{\nabla \mathbf{f}}(\mathbf{X}^{k+1}) - \widetilde{\nabla \mathbf{f}}(\mathbf{X}^k)\|_F + (1 - \beta) \|\tilde{\Delta}^k\|_F. \quad (4.74)$$

Unrolling (4.74) and averaging over $k = 0, \dots, K - 1$ gives

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\tilde{\mathbf{E}}^k\|_F \leq \frac{\mathbb{E} \|\tilde{\mathbf{E}}^0\|_F}{K(1 - \beta)} + \frac{1}{K(1 - \beta)} \sum_{k=0}^{K-1} \mathbb{E} \|\widetilde{\nabla \mathbf{f}}(\mathbf{X}^{k+1}) - \widetilde{\nabla \mathbf{f}}(\mathbf{X}^k)\|_F + \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\tilde{\Delta}^k\|_F,$$

where we used the weight identity $(1 - \beta) \sum_{s=0}^{\infty} \beta^s = 1$.

Next, since each f_i is L_* -smooth in the sense of Assumption 2, we have for each i and any X, Y that $\|\nabla f_i(X) - \nabla f_i(Y)\|_F \leq \|\nabla f_i(X) - \nabla f_i(Y)\|_* \leq L_* \|X - Y\| \leq L_* \|X - Y\|_F$. Stacking and using non-expansiveness of $(I - J)$ yields

$$\|\widetilde{\nabla \mathbf{f}}(\mathbf{X}^{k+1}) - \widetilde{\nabla \mathbf{f}}(\mathbf{X}^k)\|_F \leq L_* \|\mathbf{X}^{k+1} - \mathbf{X}^k\|_F.$$

Moreover, by Assumption 3 and Jensen, $\mathbb{E} \|\tilde{\Delta}^k\|_F \leq \mathbb{E} \|\Delta^k\|_F \leq \sigma \sqrt{N}$. Therefore,

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\tilde{\mathbf{E}}^k\|_F \leq \frac{\mathbb{E} \|\tilde{\mathbf{E}}^0\|_F}{K(1 - \beta)} + \frac{L_*}{1 - \beta} \cdot \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\mathbf{X}^{k+1} - \mathbf{X}^k\|_F + \sigma \sqrt{N}. \quad (4.75)$$

Finally, note that $\tilde{\mathbf{M}}^{k+1} - \tilde{\mathbf{M}}^k = (1 - \beta) (\tilde{\mathbf{G}}^k - \tilde{\mathbf{M}}^k) = (1 - \beta) (\tilde{\Delta}^k + \tilde{\mathbf{E}}^k)$. Thus, using triangle inequality and (4.75),

$$\begin{aligned} \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\tilde{\mathbf{M}}^{k+1} - \tilde{\mathbf{M}}^k\|_F &\leq (1 - \beta) \cdot \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\tilde{\Delta}^k\|_F + (1 - \beta) \cdot \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\tilde{\mathbf{E}}^k\|_F \\ &\leq (1 - \beta) \sigma \sqrt{N} + \frac{\mathbb{E} \|\tilde{\mathbf{E}}^0\|_F}{K} + L_* \cdot \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\mathbf{X}^{k+1} - \mathbf{X}^k\|_F + (1 - \beta) \sigma \sqrt{N}. \end{aligned}$$

It remains to bound the average primal increment. Write $\mathbf{X}^k = \bar{\mathbf{X}}^k + \tilde{\mathbf{X}}^k$ with $\bar{\mathbf{X}}^k := J\mathbf{X}^k$. Then

$$\|\mathbf{X}^{k+1} - \mathbf{X}^k\|_F \leq \|\bar{\mathbf{X}}^{k+1} - \bar{\mathbf{X}}^k\|_F + \|\tilde{\mathbf{X}}^{k+1} - \tilde{\mathbf{X}}^k\|_F.$$

Since $\bar{\mathbf{X}}^k$ is a stacked consensus variable with all blocks equal to $\bar{X}^k$, we have $\|\bar{\mathbf{X}}^{k+1} - \bar{\mathbf{X}}^k\|_F = \sqrt{N} \|\bar{X}^{k+1} - \bar{X}^k\|_F$. By Lemma 4.2, $\|\bar{X}^{k+1} - \bar{X}^k\|_F \leq \alpha \sqrt{r_{\max}}$. Moreover, $\|\tilde{\mathbf{X}}^{k+1} - \tilde{\mathbf{X}}^k\|_F \leq \|\tilde{\mathbf{X}}^{k+1}\|_F + \|\tilde{\mathbf{X}}^k\|_F$. Averaging over $k=0, \dots, K-1$ yields

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\mathbf{X}^{k+1} - \mathbf{X}^k\|_F \leq \alpha \sqrt{Nr_{\max}} + \frac{2}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\tilde{\mathbf{X}}^k\|_F + \frac{1}{K} \mathbb{E} \|\tilde{\mathbf{X}}^K\|_F.$$

Substituting into the previous display gives (4.71).

Step 3: plug into the GT bound and (optionally) use Lemma 4.7. Combining Step 1 and Step 2 yields (4.72). If Lemma 4.7 holds with Muon directions, then $\|\tilde{\mathbf{X}}^k\|_F \leq v_1 v_2 \gamma^k (1 + \|B^2\|) \|\tilde{\mathbf{X}}^0\|_F + \frac{2\alpha v_1 v_2 \lambda_a}{1-\gamma} \sqrt{Nr_{\max}}$. Averaging and using $\frac{1}{K} \sum_{k=0}^{K-1} \gamma^k \leq \frac{1}{K(1-\gamma)}$ gives

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\tilde{\mathbf{X}}^k\|_F \leq \frac{v_1 v_2 (1 + \|B^2\|)}{K(1-\gamma)} \|\tilde{\mathbf{X}}^0\|_F + \frac{2\alpha v_1 v_2 \lambda_a}{1-\gamma} \sqrt{Nr_{\max}},$$

which implies (4.73). □

Proposition 4.4 (Transient complexity with explicit (fast) network-dependent decay). *Assume Assumptions 1–4. Let the iterates be generated by (3.1) with initialization $\mathbf{Y}^0 = 0$, $\mathbf{M}^0 = \mathbf{G}^0$, and $\mathbf{H}^0 = \mathbf{M}^0$ as in Algorithm 1. Assume also the stability condition in Lemma 4.7 so that its constants $(\gamma, v_1, v_2, \lambda_a)$ exist. Fix a horizon $K \geq 1$ and choose*

$$\alpha := \frac{\alpha_0}{K^{3/4}}, \quad 1 - \beta := \frac{b_0}{\sqrt{K}}, \quad \text{with fixed constants } \alpha_0 > 0, b_0 \in (0, 1].$$

Let $r_{\max} := \min\{m, n\}$ and $\Delta_0 := \mathbb{E}[f(\bar{X}^0)] - f_{\inf}$. Let $\lambda := \rho(W - J) \in (0, 1)$. Define the average tracking error

$$E_{\text{gt}}^k := \bar{H}^k - \nabla f(\bar{X}^k), \quad \bar{H}^k := \frac{1}{N} \sum_{i=1}^N H_i^k, \quad \bar{X}^k := \frac{1}{N} \sum_{i=1}^N X_i^k.$$

Then

$$\begin{aligned}
\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\nabla f(\bar{X}^k)\|_* &\leq \underbrace{\left(\frac{\Delta_0}{\alpha_0} + \frac{2\sigma\sqrt{r_{\max}}}{\sqrt{N}} \sqrt{b_0} + \frac{2L_*\alpha_0 r_{\max}}{b_0} \right)}_{\text{topology-free leading term}} K^{-1/4} \\
&+ \underbrace{\left(\frac{5}{2} L_* \alpha_0 + \frac{4L_*\alpha_0 v_1 v_2 \lambda_a}{1-\gamma} r_{\max} + \frac{2\lambda}{1-\lambda} L_* \alpha_0 r_{\max} + \frac{8\lambda}{1-\lambda} \frac{L_* \alpha_0 v_1 v_2 \lambda_a}{1-\gamma} r_{\max} \right)}_{\text{network-dependent drift terms (no } \sigma \text{)}} K^{-3/4} \\
&+ \underbrace{\left(\frac{2\sqrt{r_{\max}}}{b_0} \mathbb{E} \|E_{\text{gt}}^0\|_F + \frac{4\lambda}{1-\lambda} b_0 \sigma \sqrt{r_{\max}} \right)}_{\text{higher-order stochastic terms}} K^{-1/2} \\
&+ \left(\frac{2L_* v_1 v_2 (1 + \|B^2\|)}{(1-\gamma)\sqrt{N}} \sqrt{r_{\max}} \|\tilde{\mathbf{X}}^0\|_F + \frac{4\lambda}{1-\lambda} \frac{L_* v_1 v_2 (1 + \|B^2\|)}{(1-\gamma)\sqrt{N}} \sqrt{r_{\max}} \|\tilde{\mathbf{X}}^0\|_F \right. \\
&\quad + \frac{2\lambda}{1-\lambda} \frac{L_* v_1 v_2 (1 + \|B^2\|)}{\sqrt{N}} \sqrt{r_{\max}} \|\tilde{\mathbf{X}}^0\|_F + \frac{2\lambda}{1-\lambda} \sqrt{\frac{r_{\max}}{N}} \mathbb{E} \|\tilde{\nabla} \mathbf{f}(\mathbf{X}^0) - \tilde{\mathbf{M}}^0\|_F \\
&\quad \left. + \frac{2\lambda}{1-\lambda} \sqrt{\frac{r_{\max}}{N}} \mathbb{E} \|\tilde{\mathbf{H}}^0\|_F \right) \underbrace{K^{-1}}_{\text{init. transients}} \\
&+ \underbrace{\frac{4\lambda}{1-\lambda} \frac{L_* \alpha_0 v_1 v_2 \lambda_a}{1-\gamma} r_{\max} K^{-7/4}}_{\text{fast transient from } \alpha/K}.
\end{aligned} \tag{4.76}$$

In particular,

$$\lim_{K \rightarrow \infty} \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\nabla f(\bar{X}^k)\|_* = 0.$$

Proof. **Step 1: start from the telescoping descent inequality.** Proposition 4.2 gives

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\nabla f(\bar{X}^k)\|_* \leq \frac{\Delta_0}{\alpha K} + \frac{L_* \alpha}{2} + \frac{2}{K} \sum_{k=0}^{K-1} \mathbb{E} \left[\frac{1}{N} \sum_{i=1}^N \|\nabla f(\bar{X}^k) - H_i^{k+1}\|_* \right]. \tag{4.77}$$

Step 2: reduce the tracking mismatch to averaged GT error and GT disagreement. For each k ,

$$\mathbb{E} \left[\frac{1}{N} \sum_{i=1}^N \|\nabla f(\bar{X}^k) - H_i^{k+1}\|_* \right] \leq L_* \alpha + \sqrt{r_{\max}} \mathbb{E} \|E_{\text{gt}}^{k+1}\|_F + \sqrt{\frac{r_{\max}}{N}} \mathbb{E} \|\tilde{\mathbf{H}}^{k+1}\|_F.$$

Averaging over $k=0, \dots, K-1$ and substituting into (4.77) yields

$$\begin{aligned} \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\nabla f(\bar{X}^k)\|_* &\leq \frac{\Delta_0}{\alpha K} + \frac{5}{2} L_* \alpha + 2\sqrt{r_{\max}} \cdot \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|E_{\text{gt}}^{k+1}\|_F \\ &\quad + 2\sqrt{\frac{r_{\max}}{N}} \cdot \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\tilde{\mathbf{H}}^{k+1}\|_F. \end{aligned} \quad (4.78)$$

Step 3: bound the averaged GT error and the averaged GT disagreement. Lemma 4.16 gives

$$\begin{aligned} \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|E_{\text{gt}}^{k+1}\|_F &\leq \sqrt{\frac{1-\beta}{1+\beta}} \frac{\sigma}{\sqrt{N}} + \frac{L_*}{1-\beta} \cdot \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\bar{X}^{k+1} - \bar{X}^k\|_F \\ &\quad + \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\bar{g}^k - \nabla f(\bar{X}^k)\|_F + \frac{1}{K} \sum_{k=0}^{K-1} \beta^{k+1} \mathbb{E} \|E_{\text{gt}}^0\|_F. \end{aligned} \quad (4.79)$$

By Lemma 4.2, $\|\bar{X}^{k+1} - \bar{X}^k\|_F \leq \alpha \sqrt{r_{\max}}$. Moreover, by (4.67) and Lemma 4.7 (Muon case),

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\bar{g}^k - \nabla f(\bar{X}^k)\|_F \leq \frac{L_*}{\sqrt{N}} \cdot \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\tilde{\mathbf{X}}^k\|_F.$$

By Lemma 4.7 with Muon directions (using (4.32)), for every $k \geq 0$,

$$\|\tilde{\mathbf{X}}^k\|_F \leq v_1 v_2 \gamma^k (1 + \|B^2\|) \|\tilde{\mathbf{X}}^0\|_F + \frac{2\alpha v_1 v_2 \lambda_a}{1-\gamma} \sqrt{N r_{\max}},$$

and hence, averaging over $k=0, \dots, K-1$ and using $\frac{1}{K} \sum_{k=0}^{K-1} \gamma^k \leq \frac{1}{K(1-\gamma)}$,

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\tilde{\mathbf{X}}^k\|_F \leq \frac{v_1 v_2 (1 + \|B^2\|)}{(1-\gamma)K} \|\tilde{\mathbf{X}}^0\|_F + \frac{2\alpha v_1 v_2 \lambda_a}{1-\gamma} \sqrt{N r_{\max}}.$$

In particular, the primal disagreement is of order $O(\alpha)$ up to an exponentially decaying/initial transient, and this bound is *independent of the gradient noise level σ* thanks to the uniform boundedness of Muon directions.

For the GT disagreement, Lemma 4.17 yields (4.72). In this bound, the stochastic-noise contribution enters multiplied by $(1-\beta)$ (via the EMA increment), giving a term of the form

$$\frac{\lambda}{1-\lambda} (1-\beta) \sigma \sqrt{N},$$

which becomes a higher-order term of order $(1-\beta)\sigma = O(K^{-1/2})$ under $1-\beta = b_0 K^{-1/2}$. The remaining deterministic contribution in (4.72) is controlled by smoothness through the primal increments and the primal disagreement; under Muon, these increments

are $O(\alpha)$. Moreover, Lemma 4.17 contains an additional boundary term of the form $\frac{L_*}{K} \mathbb{E} \|\tilde{\mathbf{X}}^K\|_F$. Using Lemma 4.7 (Muon case) at time $k=K$ and $\gamma^K \leq 1$ yields

$$\frac{1}{K} \mathbb{E} \|\tilde{\mathbf{X}}^K\|_F \leq \frac{v_1 v_2 (1 + \|B^2\|)}{K} \|\tilde{\mathbf{X}}^0\|_F + \frac{2\alpha v_1 v_2 \lambda_a}{(1-\gamma)K} \sqrt{Nr_{\max}}.$$

The first term contributes an additional K^{-1} initial transient, while the second contributes a faster term of order α/K . Substituting the GT disagreement bound (4.72) (together with the averaged primal-disagreement bound above) into (4.78) yields the claimed K -dependent decomposition.

Step 4: substitute $\alpha = \alpha_0 K^{-3/4}$ **and** $1 - \beta = b_0 K^{-1/2}$. Collect powers of K using

$$\begin{aligned} \frac{\Delta_0}{\alpha K} &= \frac{\Delta_0}{\alpha_0} K^{-1/4}, & \sqrt{1-\beta} &= \sqrt{b_0} K^{-1/4}, & \frac{\alpha}{1-\beta} &= \frac{\alpha_0}{b_0} K^{-1/4}, \\ \frac{1}{K} \sum_{k=0}^{K-1} \beta^{k+1} &\leq \frac{1}{K(1-\beta)} = \frac{1}{b_0} K^{-1/2}, & \alpha &= \alpha_0 K^{-3/4}, & \frac{\alpha}{K} &= \alpha_0 K^{-7/4}, \\ 1-\beta &= b_0 K^{-1/2}, & \frac{1}{K} &= K^{-1}. \end{aligned}$$

This yields (4.76). □

Corollary 4.1 (Big-O form of the SUDA–Muon convergence rate). *Under the assumptions of Proposition 4.4 and with the same stepsize and EMA choices $\alpha = \alpha_0 K^{-3/4}$ and $1 - \beta = b_0 K^{-1/2}$, there exists a constant $C > 0$ (independent of K, N, λ, γ , and σ) such that for all integers $K \geq 1$,*

$$\begin{aligned} \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\bar{X}^k)\|_*] &\leq C \left(\underbrace{\left(1 + \frac{\sigma}{\sqrt{N}}\right) K^{-1/4}}_{\text{dominant topology-free term}} + \underbrace{\frac{K^{-3/4}}{(1-\gamma)(1-\lambda)}}_{\text{network-dependent drift terms}} \right. \\ &\quad + \underbrace{\left(1 + \frac{\sigma}{1-\lambda}\right) K^{-1/2}}_{\text{higher-order stochastic network term}} + \underbrace{\frac{K^{-1}}{\sqrt{N}(1-\gamma)(1-\lambda)}}_{\text{network-dependent initial transient}} \\ &\quad \left. + \underbrace{\frac{K^{-7/4}}{(1-\gamma)(1-\lambda)}}_{\text{fast transient}} \right). \end{aligned} \quad (4.80)$$

Equivalently,

$$\begin{aligned} \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\bar{X}^k)\|_*] &= \mathcal{O} \left(\left(1 + \frac{\sigma}{\sqrt{N}}\right) K^{-1/4} + \frac{K^{-3/4}}{(1-\gamma)(1-\lambda)} \right. \\ &\quad \left. + \left(1 + \frac{\sigma}{1-\lambda}\right) K^{-1/2} + \frac{K^{-1}}{\sqrt{N}(1-\gamma)(1-\lambda)} + \frac{K^{-7/4}}{(1-\gamma)(1-\lambda)} \right). \end{aligned}$$

Proof. Starting from the refined bound (4.76) in Proposition 4.4, note that it decomposes into five groups of terms with powers $K^{-1/4}$, $K^{-3/4}$, $K^{-1/2}$, K^{-1} , and $K^{-7/4}$. We upper-bound each coefficient by a universal constant times the desired $(K, N, 1-\lambda, 1-\gamma, \sigma)$ -dependent factor.

1. $K^{-1/4}$ term. The coefficient of $K^{-1/4}$ in (4.76) is

$$\frac{\Delta_0}{\alpha_0} + \frac{2\sigma\sqrt{r_{\max}}}{\sqrt{N}} \sqrt{b_0} + \frac{2L_*\alpha_0 r_{\max}}{b_0},$$

which is bounded by a constant depending only on $(\Delta_0, L_*, r_{\max}, \alpha_0, b_0)$ times $1 + \sigma/\sqrt{N}$. This yields the first term in (4.80).

2. $K^{-3/4}$ term. The $K^{-3/4}$ coefficient in (4.76) contains only deterministic drift terms and depends on γ and λ via expressions of the form $1/(1-\gamma)$ and $\lambda/(1-\lambda)$. Using $0 < \gamma < 1$ and $0 < \lambda < 1$, we have $1 \leq 1/(1-\gamma)$ and $\lambda/(1-\lambda) \leq 1/(1-\lambda)$, so the entire coefficient is bounded by a constant multiple of $1/((1-\gamma)(1-\lambda))$. This gives the second term in (4.80).

3. $K^{-1/2}$ term. The $K^{-1/2}$ group in (4.76) consists of

$$\frac{2\sqrt{r_{\max}}}{b_0} \mathbb{E} \|E_{\text{gt}}^0\|_F + \frac{4\lambda}{1-\lambda} b_0 \sigma \sqrt{r_{\max}},$$

which we bound by a constant (independent of $K, N, \lambda, \gamma, \sigma$) plus a constant multiple of $\sigma/(1-\lambda)$. Absorbing the purely deterministic part into C yields the third term in (4.80).

4. K^{-1} term. The K^{-1} group in (4.76) collects initial transients. Factoring out the explicit $\sqrt{r_{\max}}/\sqrt{N}$ and using again $1 \leq 1/(1-\gamma)$ and $\lambda/(1-\lambda) \leq 1/(1-\lambda)$, the whole coefficient is bounded by a constant multiple of $1/(\sqrt{N}(1-\gamma)(1-\lambda))$, giving the fourth term in (4.80).

5. $K^{-7/4}$ term. The last group in (4.76) is already of the form $\text{const} \cdot \lambda/((1-\lambda)(1-\gamma)) K^{-7/4}$. Using $\lambda \leq 1$, we upper-bound this by a constant times $K^{-7/4}/((1-\gamma)(1-\lambda))$, which yields the fifth term in (4.80).

Combining these five estimates and enlarging C if necessary, we obtain (4.80). The equivalent big- $\mathcal{O}$ expression follows immediately. $\square$

With the master bound established, we now draw two structural consequences. We first instantiate it for concrete SUDA communication backbones in Subsection 4.4 and then show in Subsection 4.5 that tracking before polarization is not a modular detail but a structural requirement. Section 5 then turns to a complementary counterexample on federated versus fully decentralized linear speedup.

4.4 Specialization to Concrete SUDA Backbones

To compare concrete communication backbones, it suffices to inspect how Corollary 4.1 depends on the SUDA contraction rate γ . For convenience, we recall the big- $\mathcal{O}$ master bound:

$$\begin{aligned} \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\bar{X}^k)\|_*] \leq C \left(\left(1 + \frac{\sigma}{\sqrt{N}}\right) K^{-1/4} + \frac{K^{-3/4}}{(1-\gamma)(1-\lambda)} \right. \\ \left. + \left(1 + \frac{\sigma}{1-\lambda}\right) K^{-1/2} + \frac{K^{-1}}{\sqrt{N}(1-\gamma)(1-\lambda)} \right. \\ \left. + \frac{K^{-7/4}}{(1-\gamma)(1-\lambda)} \right), \end{aligned} \quad (4.81)$$

The dominant term is the same for every SUDA instantiation and is completely independent of the graph. Different communication backbones appear only through the factor $(1-\gamma)^{-1}$ in the lower-order network corrections, where γ is the SUDA contraction rate.

For the SUDA instantiations of theoretical interest here, the substitution is simple:

$$\gamma = \lambda^2 \quad \text{for ED/D}^2 \text{ and EXTRA}, \quad \gamma = \lambda \quad \text{for ATC-GT.}$$

Therefore

$$\frac{1}{(1-\gamma)(1-\lambda)} = \mathcal{O}\left(\frac{1}{(1-\lambda)^2}\right)$$

in all three cases. At the big- $\mathcal{O}$ level, ED/D², EXTRA, and ATC-GT therefore all scale as $\mathcal{O}((1-\lambda)^{-2})$ in the $K^{-3/4}$, K^{-1} , and $K^{-7/4}$ network-dependent corrections.

The difference lies in constants rather than order. ED/D² and EXTRA use $1-\gamma = (1-\lambda)(1+\lambda)$, whereas ATC-GT uses $1-\gamma = 1-\lambda$. When λ is close to one, ATC-GT therefore carries a larger prefactor in the subdominant network terms, even though the asymptotic order is the same. This is precisely the modularity provided by the SUDA formulation: among convergent SUDA instantiations, the leading stochastic term is graph-free, and the communication backbone changes only the transient network corrections through the substitution for γ .

In short, the theory does not predict different dominant rates for ED/D², EXTRA, and ATC-GT. It predicts the same leading order and different transient behavior. This modularity, however, has a clear limit: changing the backbone within SUDA affects only transient terms, but changing the order between tracking and polarization does not. The next subsection shows that moving tracking after polarization can destroy convergence altogether.

4.5 Why Tracking Must Precede Polarization

The previous subsection treated the communication backbone as a modular choice inside the SUDA template. We now turn to a different design choice that is not modular in this sense: whether Muon polarizes a tracked quantity or a purely local one. A key design choice in SUDA–Muon is that the Muon direction is computed from a *network-aggregated* / *tracked* quantity H_i^{k+1} rather than directly from a local stochastic gradient (or its EMA). The following proposition shows that if we remove this pre-polarization tracking step while keeping the rest of the algorithm unchanged, then one cannot, in general, guarantee *exact* convergence to stationarity for heterogeneous objectives. For clarity we present the obstruction in the deterministic/noiseless subcase $\sigma=0$; this is *not* an additional assumption, since the stochastic-oracle model in Assumption 3 explicitly allows $\sigma=0$ (deterministic gradients), and any convergence guarantee under that model must hold in particular for this simplest admissible case. In this setting, we construct smooth and lower-bounded problems on which the averaged iterate $\bar{X}^k$ gets stuck at a *non-stationary* point.

Proposition 4.5 (A non-stationary fixed point for the no-tracking variant (general matrix case)). *For notational simplicity, consider the deterministic/noiseless setting $\sigma=0$ and the general matrix case $m, n \geq 1$. Define the no-tracking variant of SUDA–Muon by keeping Algorithm 1 unchanged except that the gradient-tracking update is removed and replaced by*

$$H_i^{k+1} \leftarrow M_i^{k+1} \quad \text{for all } i \in [N], k \geq 0. \quad (4.82)$$

Assume the communication/SUDA matrices satisfy Assumption 1, and initialize as in Algorithm 1 with $Y_i^0 = 0$ and $M_i^0 = G_i^0$.

Then there exist $N=2$ and local objectives $f_1, f_2: \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ that are L_ -smooth in the sense of Assumption 2 and are lower bounded, such that for the consensus initialization $X_1^0 = X_2^0 = X^\dagger$, the averaged iterate satisfies*

$$\bar{X}^k \equiv X^\dagger \quad \text{for all } k \geq 0,$$

while the global objective $f(X) \triangleq \frac{1}{2}(f_1(X) + f_2(X))$ has

$$\nabla f(X^\dagger) \neq 0.$$

Consequently,

$$\frac{1}{K} \sum_{k=0}^{K-1} \|\nabla f(\bar{X}^k)\|_* \equiv \|\nabla f(X^\dagger)\|_* > 0,$$

and hence the no-tracking variant (4.82) cannot guarantee exact stationarity convergence (vanishing averaged gradient nuclear norm at $\bar{X}^k$) over the class of smooth heterogeneous objectives.

Proof. We first note that the averaging argument in the proof of Lemma 4.2 uses only the $\mathbf{X}$ - and $\mathbf{Y}$ -updates (and the doubly-stochasticity / polynomial properties of A, B, C), and

does not rely on the specific recursion defining $\mathbf{H}^{k+1}$. Hence the same argument applies to the no-tracking variant as well and yields

$$\bar{X}^{k+1} = \bar{X}^k - \alpha \bar{S}^{k+1}, \quad \bar{S}^{k+1} \triangleq \frac{1}{2} \sum_{i=1}^2 S_i^{k+1}, \quad S_i^{k+1} = \text{msgn}(H_i^{k+1}). \quad (4.83)$$

Under the no-tracking rule (4.82), we have $H_i^{k+1} = M_i^{k+1}$. It remains to exhibit a smooth lower-bounded instance for which $\bar{S}^{k+1} \equiv 0$ while $\nabla f(\bar{X}^0) \neq 0$.

Construction of the counterexample. Choose f_1, f_2 as in Example 4.1 below and initialize at consensus $\bar{X}^0 = X^\dagger$. For these functions, for every $X \in \mathbb{R}^{m \times n}$, the gradients satisfy

$$\nabla f_1(X) = c_1(X)U, \quad \nabla f_2(X) = c_2(X)U, \quad (4.84)$$

where $U \in \mathbb{R}^{m \times n}$ is a fixed rank-one matrix and the scalar coefficients obey $c_1(X) > 0$ and $c_2(X) < 0$. In particular, writing $U = uv^\top$ with $\|u\|_2 = \|v\|_2 = 1$, for any scalar $\mu \neq 0$ we have $\text{msgn}(\mu U) = \text{sign}(\mu)U$ because a reduced SVD of μU is $\mu U = u(|\mu|)(\text{sign}(\mu)v)^\top$. Therefore, $\text{msgn}(\nabla f_1(X)) = U$ and $\text{msgn}(\nabla f_2(X)) = -U$ for all X .

Inductive cancellation argument. In the noiseless setting, $G_i^k = \nabla f_i(X_i^k)$. Since $M_i^0 = G_i^0$ and $M_i^{k+1} = \beta M_i^k + (1 - \beta)G_i^k$, the alignment (4.84) implies inductively that $M_1^k = \mu_1^k U$ and $M_2^k = \mu_2^k U$ for scalars $\mu_1^k > 0$ and $\mu_2^k < 0$ for all k . Therefore, under (4.82), we have $S_1^{k+1} = \text{msgn}(M_1^{k+1}) = U$ and $S_2^{k+1} = \text{msgn}(M_2^{k+1}) = -U$, and hence $\bar{S}^{k+1} = (U - U)/2 = 0$ for all $k \geq 0$. Plugging $\bar{S}^{k+1} = 0$ into (4.83) yields $\bar{X}^{k+1} = \bar{X}^k$ for all k , so $\bar{X}^k \equiv \bar{X}^0 = X^\dagger$. Finally, Example 4.1 satisfies $\nabla f(X^\dagger) \neq 0$, so the averaged stationarity measure stays bounded away from zero. $\square$

Example 4.1 (A concrete smooth counterexample in the general matrix setting). Let $N=2$. Pick any unit vectors $u \in \mathbb{R}^m$ and $v \in \mathbb{R}^n$ with $\|u\|_2 = \|v\|_2 = 1$ and define the rank-one matrix

$$U \triangleq uv^\top.$$

Let $a > b > 0$ be constants and define, for $X \in \mathbb{R}^{m \times n}$,

$$f_1(X) \triangleq a \log(1 + e^{\langle U, X \rangle}), \quad f_2(X) \triangleq b \log(1 + e^{-\langle U, X \rangle}),$$

where $\langle U, X \rangle \triangleq \text{trace}(U^\top X)$. Then f_1 and f_2 are lower bounded (both are nonnegative) and differentiable with

$$\nabla f_1(X) = a \frac{1}{1 + e^{-\langle U, X \rangle}} U, \quad \nabla f_2(X) = -b \frac{1}{1 + e^{\langle U, X \rangle}} U.$$

In particular, $\nabla f_1(X)$ and $\nabla f_2(X)$ are always aligned with U but have opposite signs in their scalar coefficients; hence $\text{msgn}(\nabla f_1(X)) = uv^\top$ and $\text{msgn}(\nabla f_2(X)) = -uv^\top$ for all X .

Moreover, the gradients are Lipschitz in the sense of Assumption 2. Indeed, let $\varphi(t) \triangleq (1+e^{-t})^{-1}$ so that $\varphi'(t) = \varphi(t)(1-\varphi(t)) \leq 1/4$. Since $\|U\|_* = 1$ (rank-one with unit singular value) and $|\langle U, X-Y \rangle| \leq \|U\|_* \|X-Y\| = \|X-Y\|$, we have

$$\|\nabla f_1(X) - \nabla f_1(Y)\|_* = a |\varphi(\langle U, X \rangle) - \varphi(\langle U, Y \rangle)| \|U\|_* \leq \frac{a}{4} |\langle U, X-Y \rangle| \leq \frac{a}{4} \|X-Y\|,$$

and similarly $\|\nabla f_2(X) - \nabla f_2(Y)\|_* \leq \frac{b}{4} \|X-Y\|$. Thus Assumption 2 holds with $L_* \triangleq \max\{a, b\}/4$.

Finally, with $X^\dagger \triangleq 0$, for the global objective $f \triangleq \frac{1}{2}(f_1 + f_2)$ we have

$$\nabla f(X^\dagger) = \frac{1}{2}(\nabla f_1(0) + \nabla f_2(0)) = \frac{1}{2}\left(\frac{a}{2} - \frac{b}{2}\right)U = \frac{a-b}{4}U \neq 0.$$

Therefore, the hypotheses and conclusion of Proposition 4.5 are satisfied in the general matrix setting.

This structural result underscores the role of the gradient-tracking step in Algorithm 1: because msgn is nonlinear, replacing a tracked quantity by a purely local one can break exact convergence under heterogeneous objectives. The next section asks a complementary question: even when averaging is exact and topology is no obstacle, does decentralized Muon retain the familiar federated linear-speedup effect?

5 Federated vs. Decentralized Linear Speedup

The previous subsection concerned exact convergence. We now turn to a complementary scaling question: can Muon exploit the usual linear-speedup effect from averaging stochastic gradients? Here the answer again depends on *when* averaging happens. In a federated or server-based architecture, gradients can be averaged first and polarized afterward. In a fully decentralized architecture, each node polarizes a local noisy gradient and only then communicates. The following toy example isolates this structural difference in the simplest possible setting.

We consider the vector case as a special instance of our matrix formulation, namely $X^k = (x_1^k, x_2^k)^\top \in \mathbb{R}^{2 \times 1}$. Let

$$f(X) \triangleq \frac{1}{2}x_1^2, \tag{5.1}$$

so that $\nabla f(X) = (x_1, 0)^\top$. At iteration k , node $i \in [N]$ observes the stochastic gradient

$$G_i^k = \nabla f(X^k) + \begin{bmatrix} 0 \\ \xi_i^k \end{bmatrix} = \begin{bmatrix} x_1^k \\ \xi_i^k \end{bmatrix}, \quad \xi_i^k \in \{+\sigma, -\sigma\}, \tag{5.2}$$

where $\{\xi_i^k\}$ are i.i.d. across i and k with $\mathbb{E}[\xi_i^k] = 0$ and $\mathbb{E}[(\xi_i^k)^2] = \sigma^2$. The noise is transverse to the signal: it acts only in the second coordinate.

Since $X^k \in \mathbb{R}^{2 \times 1}$, the matrix-sign operator reduces to Euclidean normalization,

$$\text{msgn}(V) = \frac{V}{\|V\|_2}, \quad \text{msgn}(0) = 0. \quad (5.3)$$

To give the decentralized method its best possible chance, we assume a complete graph with exact averaging and synchronized iterates. The only distinction is the order of averaging and polarization.

$$\textbf{Decentralized Muon:} \quad X^{k+1} = X^k - \alpha \cdot \frac{1}{N} \sum_{i=1}^N \text{msgn}(G_i^k), \quad (5.4)$$

$$\textbf{Federated Muon:} \quad X^{k+1} = X^k - \alpha \cdot \text{msgn} \left(\frac{1}{N} \sum_{i=1}^N G_i^k \right). \quad (5.5)$$

Proposition 5.1 (Federated Muon can speed up, decentralized Muon cannot). *Consider (5.1)–(5.3) with any $\sigma > 0$ and stepsize $\alpha \in (0, \sigma)$.*

1. **Decentralized Muon is insensitive to N .** *The first coordinate of (5.4) obeys the deterministic recursion*

$$x_1^{k+1} = x_1^k \left(1 - \frac{\alpha}{\sqrt{(x_1^k)^2 + \sigma^2}} \right). \quad (5.6)$$

Hence

$$|x_1^{k+1}| \geq \left(1 - \frac{\alpha}{\sigma} \right) |x_1^k|,$$

and reaching $|x_1^T| \leq \varepsilon |x_1^0|$ requires

$$T = \Omega \left(\frac{\sigma}{\alpha} \log \frac{1}{\varepsilon} \right),$$

which does not improve with N .

2. **Federated Muon enjoys variance reduction.** *Let $\bar{\xi}^k \triangleq \frac{1}{N} \sum_{i=1}^N \xi_i^k$, so that $\mathbb{E}[\bar{\xi}^k] = 0$ and $\mathbb{E}[(\bar{\xi}^k)^2] = \sigma^2 / N$. Then the federated update (5.5) is driven by noise at scale $\sigma / \sqrt{N}$. In the noise-dominated regime $|x_1^k| \lesssim \sigma / \sqrt{N}$, the corresponding iteration complexity scales as*

$$T = O \left(\frac{\sigma}{\alpha \sqrt{N}} \log \frac{1}{\varepsilon} \right),$$

which is the usual linear-speedup behavior.

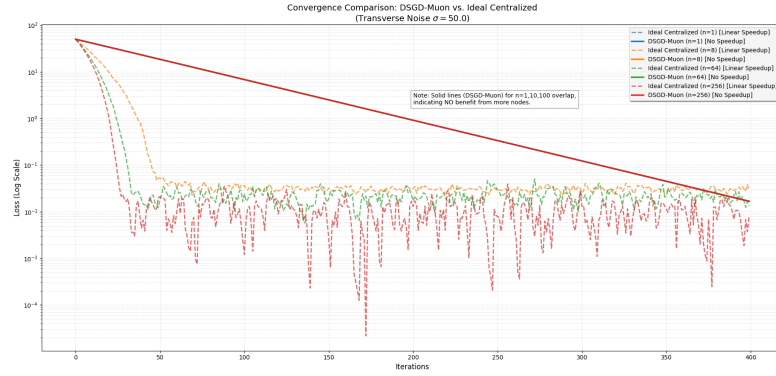


Figure 1: Counterexample for linear speedup under transverse noise ($\sigma = 50$). Dashed curves: federated Muon (5.5) with server-side averaging before polarization. Solid curves: decentralized Muon (5.4) with local polarization before averaging. As N increases, the federated curves accelerate clearly, whereas the decentralized curves essentially overlap, indicating the absence of linear speedup.

Proof sketch. For decentralized Muon, note that

$$\|G_i^k\|_2 = \sqrt{(x_1^k)^2 + (\bar{\zeta}_i^k)^2} = \sqrt{(x_1^k)^2 + \sigma^2}.$$

Therefore the first coordinate of the local Muon direction equals $x_1^k / \sqrt{(x_1^k)^2 + \sigma^2}$, which is independent of both the node index i and the sign of the noise. Averaging over i yields (5.6), and the lower bound follows from $\sqrt{(x_1^k)^2 + \sigma^2} \geq \sigma$.

For federated Muon, averaging occurs before normalization, so the second coordinate becomes $\bar{\zeta}^k$ with variance σ^2/N . The effective noise level is therefore reduced from σ to $\sigma/\sqrt{N}$, which gives the stated complexity in the noise-dominated regime. $\square$

This counterexample is useful precisely because topology plays no role in it: the graph is complete and averaging is exact. The failure of linear speedup comes solely from local polarization before averaging. In that sense, the example isolates a structural difference between federated Muon and fully decentralized Muon rather than a weakness of any particular communication scheme.

Figure 1 visualizes the same phenomenon for $\sigma=50$ and $N \in \{1, 8, 64, 256\}$. The dashed federated curves accelerate as N grows, while the solid decentralized curves nearly overlap across all choices of N .

6 Numerical Experiments

6.1 Experimental Setup

We use two benchmarks that stress different parts of the decentralized Muon story: CIFAR-100 with strong cross-node heterogeneity, and GPT-2 fine-tuning on Wikitext-2 with a much milder, near-IID partition. Throughout, we compare SUDA-MUON-ED, SUDA-MUON-ATC-GT, and DEMUON, and all quantitative summaries are computed from pre-aggregated CSV logs produced by the training scripts.

For CIFAR-100, we use the standard 50,000/10,000 train/test split, a ResNet-18-style model with 100 output classes, and a Dirichlet partition with concentration $\alpha = 0.05$, which produces severe label skew across workers. The main comparison uses $N = 10$ nodes on a ring with mixing parameter $\rho = 0.25$, batch size 512 per node, constant step-size $\eta = 0.03$, EMA parameter $\beta = 0.9$, weight decay 5×10^{-4} , and 25 training epochs. Unless otherwise stated, each configuration is run with 3 random seeds, and we report the performance of the averaged model.

We also include two CIFAR-100 extensions aimed at the communication story. First, fixing SUDA-MUON-ED, we vary the topology and node count over $N \in \{4, 8, 12\}$ to isolate how system scale changes the preferred graph. Second, we run a larger and longer experiment with $N = 20$ ring-connected nodes for 100 epochs using a slightly smaller constant stepsize $\eta = 0.01$. This long-horizon setting is where steady-state communication effects should be easiest to see.

For GPT-2/Wikitext-2, each node hosts the pretrained gpt2 model. We tokenize the corpus into length-512 sequences, use local batch size 16, train for 4 epochs with base stepsize $\eta = 0.03$, and apply a warmup-plus-cosine schedule. The main comparison uses a 15-node ring, while the ablation again varies topology and node count for SUDA-MUON-ED. Each GPT configuration is run with 2 seeds, and we report validation perplexity at epoch 4.

6.2 Results and Analysis

CIFAR-100 main comparison (10-node ring). The short-horizon 10-node ring benchmark is relatively balanced. After 25 epochs, all three decentralized Muon variants finish within about 0.6 accuracy points of one another, as shown in Table 1. SUDA-MUON-ATC-GT achieves the highest mean final accuracy, while DEMUON is statistically very close and SUDA-MUON-ED remains competitive despite the strongly non-IID split. So the right reading of this table is not that one method wins decisively at a short budget; it is that the SUDA-based instantiations already match DEMUON in the default heterogeneous ring regime.

The variance pattern is also informative. SUDA-MUON-ATC-GT has the smallest spread across seeds, which is consistent with the intuition that explicit correction of cross-node drift can stabilize training even when the final means are close. In other words, the

Table 1: CIFAR-100 top-1 test accuracy under a 10-node ring topology with Dirichlet heterogeneity $\alpha=0.05$. We report mean $\pm$ population standard deviation over three seeds for the final accuracy at epoch 25. Higher is better.

Method	#Nodes	Topology	α	ρ	Final acc (%)
SUDA-MUON-ED	10	ring	0.05	0.25	44.99 $\pm$ 5.19
SUDA-MUON-ATC-GT	10	ring	0.05	0.25	45.56$\pm$3.18
DEMUON	10	ring	0.05	0.25	45.50 $\pm$ 5.57

Table 2: CIFAR-100 test accuracy for SUDA-MUON-ED under varying numbers of nodes and topologies. We report mean $\pm$ population standard deviation over three seeds of final and best test accuracy. Bold values indicate the best performance within each node-count group.

#Nodes	Topology	Final acc (%)	Best acc (%)
4	fully connected	51.71 $\pm$ 0.94	53.71 $\pm$ 0.57
4	line	51.76 $\pm$ 1.04	53.22 $\pm$ 0.95
4	ring	53.23$\pm$0.30	55.27$\pm$1.53
4	star	52.93 $\pm$ 0.73	53.21 $\pm$ 0.33
8	fully connected	47.69 $\pm$ 0.55	47.82 $\pm$ 0.72
8	line	45.91 $\pm$ 1.27	46.07 $\pm$ 1.03
8	ring	45.90 $\pm$ 0.97	47.04 $\pm$ 0.69
8	star	50.19$\pm$1.60	50.94$\pm$0.98
12	fully connected	43.62$\pm$2.86	45.14$\pm$3.43
12	line	37.26 $\pm$ 5.32	39.40 $\pm$ 4.93
12	ring	41.72 $\pm$ 3.73	43.03 $\pm$ 4.47
12	star	42.19 $\pm$ 4.76	42.19 $\pm$ 4.76

short-horizon experiment does not yet separate the methods strongly by final accuracy, but it already suggests that the SUDA coupling is not paying an optimization penalty for using the more structured primal-dual backbone.

CIFAR-100 ablation on topology and node count. Fixing the optimizer to SUDA-MUON-ED, Table 2 shows that the preferred topology changes with scale. At $N=4$, ring is best. At $N=8$, star becomes strongest. At $N=12$, fully connected is best and the line graph degrades the most. This is the clearest finite-time sign that graph effects are real but problem dependent: the communication backbone is not a mere implementation detail, yet neither is there a single topology that wins uniformly.

This ablation is useful for the paper’s story because it isolates the SUDA backbone itself. Once the local Muon geometry is fixed, changing only the graph still produces

Table 3: CIFAR-100 training loss and test accuracy after 100 epochs on a 20-node ring topology. We report mean $\pm$ population standard deviation over three seeds. Bold entries indicate the best values.

Method	#Nodes	Epochs	Final train loss	Final test acc (%)	Seeds
SUDA-MUON-ED	20	100	0.281 ± 0.018	52.89 ± 0.82	3
SUDA-MUON-ATC-GT	20	100	0.329 ± 0.008	48.58 ± 0.73	3
DeMUON	20	100	0.386 ± 0.009	43.94 ± 1.68	3

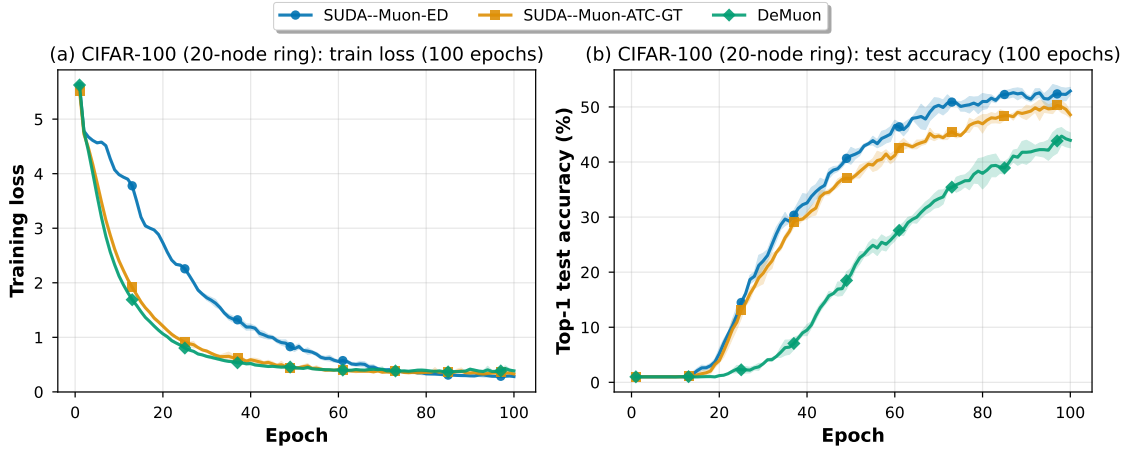


Figure 2: CIFAR-100 performance of the three decentralized Muon variants on a 20-node ring over 100 epochs. SUDA-MUON-ED converges faster and finishes higher than SUDA-MUON-ATC-GT and DEMUON. Shaded regions, where visible, denote one standard deviation over three seeds.

visibly different transient outcomes, especially as N grows. That is exactly the regime where the topology-dependent correction terms should matter most in practice, even if the leading stochastic term in the theory is topology separated.

CIFAR-100 extended training on a 20-node ring. The strongest empirical evidence for the SUDA+Muon design appears in the larger and longer non-IID run. Table 3 and Figure 2 show that SUDA-MUON-ED clearly separates from both alternatives after 100 epochs on a 20-node ring: it attains the lowest final training loss and the highest final test accuracy, outperforming SUDA-MUON-ATC-GT by more than 4 points and DEMUON by nearly 9 points in final accuracy. This is the setting where the advantage of the SUDA-style correction is no longer subtle.

The trajectory plot sharpens the comparison. During the first half of training, both SUDA-based variants already stay ahead of DEMUON; later on, SUDA-MUON-ED continues improving and cleanly overtakes SUDA-MUON-ATC-GT. So the most defensi-

Table 4: Validation perplexity on Wikitext-2 for GPT-2 under a 15-node ring topology after four training epochs. Values are means over two seeds. Lower is better; bold indicates the lowest perplexity.

Method	#Nodes	Topology	Seeds	Eval ppl (4 epochs)
SUDA-MUON-ED	15	ring	2	18.1396
SUDA-MUON-ATC-GT	15	ring	2	18.1406
DEMUON	15	ring	2	18.1401

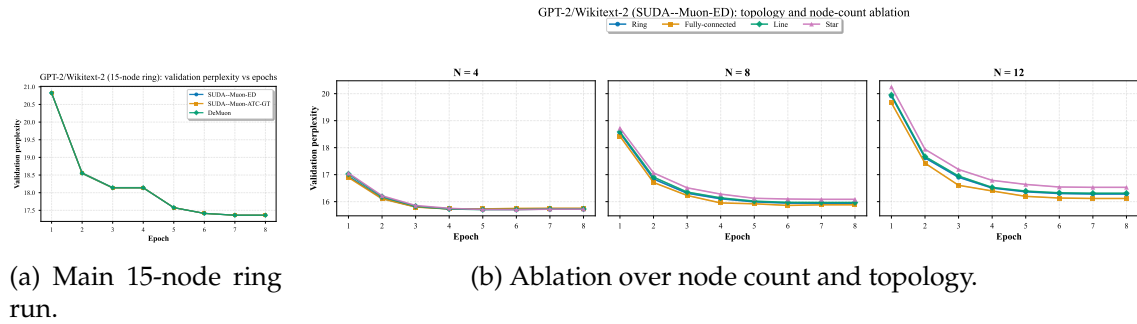


Figure 3: Validation perplexity of decentralized Muon variants on GPT-2 / Wikitext-2. The left panel compares SUDA-MUON-ED, SUDA-MUON-ATC-GT, and DEMUON under the main 15-node ring configuration, while the right panel reports the SUDA-MUON-ED topology ablation. Shaded regions denote one standard deviation over two seeds.

ble claim is tied to this specific regime: in the demanding 20-node, 100-epoch non-IID setting, both SUDA-based variants outperform DEMUON, and among the instantiations tested here SUDA-MUON-ED is the strongest.

GPT-2 / Wikitext-2 main comparison (15-node ring). The language-modeling picture is intentionally different. Under the 15-node ring, the three decentralized Muon variants are almost indistinguishable after 4 epochs, with validation perplexities clustered between 18.1396 and 18.1406 in Table 4. We therefore do not read this benchmark as evidence that one decentralized Muon mechanism dominates the others. The useful message is instead that the SUDA-based variants remain fully competitive in the benign near-IID regime; adopting the SUDA backbone does not sacrifice performance when the communication problem is mild.

GPT-2 / Wikitext-2 ablation on topology and node count. Fixing the optimizer to SUDA-MUON-ED, Table 5 shows that all topologies remain close at $N=4$, while fully connected becomes best at $N=8$ and $N=12$. The absolute spread is still small, so this is not a large-effect benchmark, but it reinforces the same practical lesson as on CIFAR-100: the graph interacts with task statistics and scale, and the preferred topology can shift as

Table 5: Validation perplexity of SUDA–MUON-ED on Wikitext-2 after four training epochs under varying numbers of nodes and topologies. Values are means over the available seeds. Lower is better; bold indicates the best performance within each node-count group.

#Nodes	Topology	Eval ppl (4 epochs)
4	fully connected	15.75
4	line	15.73
4	ring	15.74
4	star	15.72
8	fully connected	15.88
8	line	15.97
8	ring	15.94
8	star	16.09
12	fully connected	16.12
12	line	16.31
12	ring	16.28
12	star	16.53

those conditions change.

7 Conclusion

This paper studies one specific obstruction in fully decentralized Muon: the polarization step is nonlinear, so it does not interact with gossip averaging in the same way as an ordinary gradient step. SUDA–Muon addresses that obstruction by placing Muon inside the SUDA primal–dual backbone, which lets us compare several decentralized correction mechanisms within one template.

The main theoretical message is that the dominant stochastic term remains topology-free, while the graph enters only through faster-decaying corrections. The main structural message is even simpler: tracking before polarization is not optional in general. Without it, one can construct smooth heterogeneous problems on which the averaged iterate stays at a non-stationary point.

The second conceptual message is a contrast with federated Muon. In our linear-speedup counterexample, server-side averaging before polarization recovers the usual $\sigma/\sqrt{N}$ variance reduction, while fully decentralized local polarization before averaging does not, even on a complete graph with exact averaging.

Our experiments on CIFAR-100 and GPT-2/Wikitext-2 are consistent with this picture. Short-horizon and near-IID runs make the decentralized Muon variants look simi-

lar, but in the 20-node, 100-epoch non-IID CIFAR-100 setting both SUDA-based variants outperform DEMUON, with SUDA–MUON-ED performing best. When the partition is near IID, the different SUDA–Muon instances behave much more similarly and the best topology can shift.

Natural next steps include communication-efficient variants with compression, sharper rates under additional structure such as PL-type conditions, online tuning of the stepsize and EMA parameters, and larger-scale experiments that test whether the same topology effects persist for bigger models and graphs.

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